

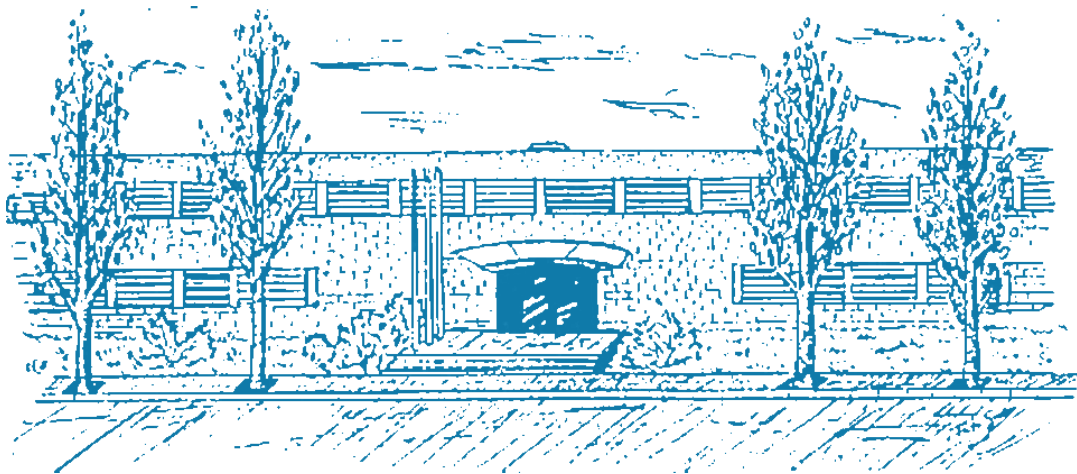
UNIVERSITAT POLITÈCNICA DE CATALUNYA - BarcelonaTech (UPC)

FME - FACULTAT DE MATEMÀTIQUES I ESTADÍSTICA

MASTER'S DEGREE IN STATISTICS AND OPERATIONS RESEARCH (MESIO UPC-UB)

MASTER'S THESIS

Uncovering latent structure in soccer data



Author: Sebastián Serra Peña

Director: Victor Peña Pizarro

Department: Departament d'Estadística i Investigació Operativa (EIO)

Barcelona, September 21, 2026



UNIVERSITAT DE
BARCELONA



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Facultat de Matemàtiques i Estadística



Universitat Politècnica de Catalunya
Facultat de Matemàtiques i Estadística

Master Thesis

Uncovering latent structure in soccer data

Sebastián Serra Peña

Advisor: Victor Peña Pizarro

September, 2026

Departament d'Estadística i Investigació Operativa (EIO)

A short dedication placed on its own page, where I have no idea who
to dedicate this too...

Abstract

Comparing football players by playing style is usually done informally, from memory and reputation, rather than from a principled statistical representation of what a player actually does on the pitch. This thesis builds a hierarchical Bayesian factor model of La Liga player-season performance to recover interpretable stylistic dimensions directly from match-event data and to measure player similarity with quantified uncertainty. The data cover 4,586 player-seasons from 1,529 outfield players across twelve seasons (2011/12–2022/23), summarised by 48 standardised performance features spanning passing, duelling, defending, and attacking output.

Each player-season observation is decomposed additively into a stable player effect, a season effect, and a residual term. The player effect follows a low-rank factor structure, and residuals follow a Student- t distribution whose scale is further adjusted for minutes played, correcting a systematic under-fit of low-minute seasons observed under a constant-variance specification. A standard identifiability constraint on the factor loadings is resolved after fitting through a rotation-invariant transformation, recovering interpretable canonical directions of style. Posterior diagnostics show excellent convergence, and the fitted covariance reproduces the observed cross-feature correlation structure closely ($r = 0.975$). A three-factor solution is retained on a qualitative variance-share and loading-reliability argument, after comparison with alternative ranks.

The three recovered factors admit a direct football reading: Defensive engagement (defensive build-up volume against attacking output), Technical orientation (ball circulation against aerial and physical duels), and Possession retention (clean possession and direct threat against wide delivery). Projecting every player onto these three axes recovers five clusters that align closely with recognisable positional roles, from forwards to centre-backs. Building on the same shared covariance structure, the thesis introduces a bounded similarity score between any two players, calibrated against the population's own typical style distance. Four case studies illustrate its behaviour: Lionel Messi is simultaneously extreme on all three axes, and even his closest style-neighbour, Neymar, reaches only 45% similarity; Sergio Ramos's closest match, Gerard Piqué, reaches 93%, consistent with their well-known on-pitch pairing; and Cristian Stuani's closest style-neighbours share his profile but not, by reputation, his output, exposing a genuine limit of a style-only measure.

The resulting framework offers a reproducible route from raw event data to a low-dimensional, uncertainty-aware representation of playing style, directly applicable to style-based scouting shortlists that complement, rather than replace, judgement on age, price, and performance.

Key words: hierarchical Bayesian models, factor analysis, latent variable models, soccer analytics, player similarity, Markov chain Monte Carlo

MSC (2020): 62F15, 62H25, 62H30

Contents

Chapter 1. Introduction	1
Chapter 2. Data source and construction	3
Chapter 3. Background	7
1. Similarity representation	8
2. Exploratory low-dimensional structure	9
Chapter 4. Model	19
1. Motivation and overview	19
2. Notation	19
3. The additive hierarchical decomposition	20
4. Covariance structure	21
5. Likelihood specification	22
6. Model identification	24
7. Prior specification	25
8. Rank selection	26
9. Variance decomposition and intraclass correlation	28
10. Canonical factor directions via spectral decomposition	29
Chapter 5. Results	31
1. Model diagnostics and convergence	31
2. Resolving the observation-level variance scaling (ϕ)	33
3. Rank selection: results	34
4. Variance decomposition: intraclass correlations	35
5. Season profiles	36
6. Canonical factor loadings by feature group	37
7. The canonical style space	39
8. Player similarity: measuring playing style	41
9. Applications: scouting and recommendations	46
10. Future work	46
Chapter 6. Conclusion	49
Appendix A. Full intraclass-correlation table	51
Appendix B. Full canonical factor loadings	53
Appendix C. SQL extraction and construction queries	59
1. Extraction and construction logic	59
Appendix. Bibliography	63

CHAPTER 1

Introduction

Modern football data analysis increasingly depends on the ability to compare players and teams in a principled, scalable manner. Clubs, analysts, and recruitment departments are often faced with questions that are conceptually simple yet technically demanding: *Which players are truly similar in style? How does a player’s role change across seasons or clubs? Can we characterise tactical profiles in a way that is both interpretable and statistically grounded?* Recent work in sports analytics and the broader literature on tactical behaviour highlights the growing importance of data-driven representations of playing style, collective dynamics, and contextual effects [24, 19]. However, many operational approaches remain either heuristic or too tightly tied to one-off visualisations, making it difficult to quantify similarity consistently across large populations, seasons, and evolving contexts.

This masters’ thesis is motivated by a practical and scientific aim: to build a framework for *modelling* latent football style from player performance data using a hierarchical Bayesian model. Rather than attempting to predict a single outcome (e.g., goals or match results), the focus is on constructing a space in which players (and, ultimately, groups such as teams or positional archetypes) can be compared meaningfully. The Bayesian approach adopted here provides, as a direct consequence rather than an added feature, posterior uncertainty on the quantities of interest — the player/season/residual variance decomposition, the canonical factor loadings, and each player’s latent position — rather than point estimates alone [23]. While Bayesian hierarchical modelling has a long track record for football outcome prediction [2] and performance modelling [20], comparatively less attention has been paid to Bayesian latent-space constructions tailored to similarity and style, particularly with hierarchical structure reflecting repeated measures across seasons.

The starting point of this work is a longitudinal dataset of player match statistics from LaLiga, covering twelve seasons (2011/12 to 2022/23) and aggregated to the player–season level. Each observation corresponds to a player in a given season, with a minimum exposure threshold to reduce noise from small samples. From these aggregates, we construct feature vectors designed to be comparable across players and seasons: volume and intensity measures expressed per 90 minutes, efficiency ratios (such as passing accuracy or duel win rates), and interpretable blocks reflecting common scouting and coaching perspectives (offensive production; progression and creation; carrying and 1v1 actions; defence and pressing; duels and physical contests). This block structure is not merely cosmetic, but rather facilitates encoding a domain-informed decomposition of performance into components that are often discussed qualitatively, and it provides a natural handle for weighting or extending the dataset when needed.

The resulting player–season observations are inherently multivariate, with substantial dependence structure across variables. Many football actions co-occur mechanically or tactically, and performance indicators often reflect overlapping dimensions of role, team context, and style. In such settings, analysing each variable in isolation can obscure the broader structure of the data, while naive comparisons in the original feature space may overstate noise, redundancy, or scale effects. A central methodological idea of this thesis is therefore to seek lower-dimensional latent representations that preserve the most relevant structure for comparison and interpretation. In practical terms, the goal is to represent player profiles in a way that is compact enough to visualise and reason about, yet rich enough to retain meaningful distinctions in style, usage, and tactical function.

From this perspective, the thesis is situated within *latent space modelling*. The main idea is that observed football statistics may be viewed as noisy, high-dimensional manifestations of a smaller number of underlying dimensions related to playing style, role execution, and tactical behaviour. In this work, factor-analytic representations provide the route used to extract interpretable latent structure from correlated performance variables. This is especially useful in a longitudinal setting, where repeated player–season observations motivate extensions with hierarchical structure and raise substantive questions about persistence, evolution, and context dependence in football style.

The overarching objectives of this master’s thesis are therefore:

- (1) to construct a robust and interpretable representation of player performance profiles at the player–season level, grounded in football-relevant blocks of execution;
- (2) to study how multivariate football data can be represented in lower-dimensional latent spaces without losing their most important stylistic and tactical information;
- (3) to investigate factor-analytic latent-variable approaches for the representation and comparison of player profiles;
- (4) to provide Bayesian posterior uncertainty for the latent-space estimation, rank selection, and variance decomposition used throughout;
- (5) to frame the resulting representations as practical tools for scouting, player profiling, and tactical interpretation in modern football analytics [24, 19].

The remainder of this work is organised as follows. Chapter 2 describes the construction of the player–season dataset, from the raw match-event data through feature engineering to the final analytical sample. Chapter 3 introduces the exploratory methodology used to establish that the resulting feature space contains meaningful lower-dimensional structure, including classical factor analysis and multidimensional scaling. Chapter 4 specifies the hierarchical Bayesian factor model used for the main analysis: its additive decomposition, covariance structure, identification strategy, and rank selection. Chapter 5 reports model diagnostics, the resulting canonical factor directions and player archetypes, and case studies on player similarity and its scouting applications, and closes with the model’s limitations and directions for future work, connecting this line of work to broader Bayesian modelling in sports [2, 20, 23].

Part of the R and Stan code supporting this thesis — model implementation, debugging, and pipeline refactoring — was developed with the assistance of an AI coding assistant (Claude, Anthropic), used under the author’s direction and review throughout. The statistical modelling decisions, the analysis, and the interpretation of results are the author’s own.

CHAPTER 2

Data source and construction

The pillars of this master’s thesis is a longitudinal player–season dataset built from Wyscout football data. Wyscout data are event-based: match actions are recorded and tagged, and later summarised into match-level player statistics [27]. The data used here were accessed through a professional Redshift warehouse in a football industry setting, with access facilitated through Sebastien Coustou at Parma Calcio. The starting point was not a ready-made player-style table, but a staged match–player table that required filtering, aggregation, and feature engineering before any modelling itself.

The scope of the dataset is deliberately restricted to historical LaLiga data. This is not intended as a deficiency of the analysis, but as a design choice. The goal of the thesis is to study latent structure over time within a single competition, thereby avoiding the additional layer of heterogeneity introduced by cross-league differences in tempo, officiating, tactical distributions, data coverage, and competitive context. Extending the framework to several leagues is possible and potentially valuable, but such an extension would constitute a modelling problem of its own, rather than a harmless enlargement of the sample.

The final analytical object is a matrix of player–season observations. Each row corresponds to one outfield player in one LaLiga season, and each column corresponds to an engineered football feature. After filtering and aggregation, the working dataset contains $N = 4586$ player–season observations, $I = 1529$ unique players, and $P = 48$ transformed football features. Player identifiers, player names, season identifiers, competition identifiers, minutes played, and matches were retained as metadata, but were not used as active variables in the feature matrix for PCA, factor analysis, dissimilarity construction, or the Bayesian model of Chapter 4.

Operationally, the starting point was a staged Redshift table at the match–player level, with one row per player and match. This table contained event-derived counts, success counts, averages, and percentages. It was joined to auxiliary competition, season, and player metadata tables in order to identify LaLiga seasons and recover player-level descriptors. The variables available at this stage included goals, assists, shots, passes, progressive passes, dribbles, duels, aerial duels, recoveries, pressing duels, touches in the box, and expected-goals-related quantities. The extraction and aggregation logic is sketched in Appendix C.

The staged data covers twelve LaLiga seasons available in the extraction, from 2011/12 through 2022/23. Goalkeeper observations were removed before constructing the similarity representation. Rather than relying only on position labels, match-level rows with goalkeeper-specific activity (shots faced, saves, conceded goals, clean sheets, or goalkeeper exits) were excluded. This restriction is methodological: goalkeepers occupy a different action space from outfield players, so including them would make part of the distance structure reflect positional incomparability rather than stylistic similarity among comparable profiles.

The remaining match–player rows were aggregated to the player–season level. Match-level quantities were summed or combined into season totals for minutes, matches, starts, goals, assists, shots, expected goals, expected assists, passes (including forward, back, lateral, long, smart, progressive, and through passes), crosses, dribbles, progressive runs, accelerations, defensive actions (interceptions, clearances, blocked shots, sliding tackles), duels (defensive, offensive, aerial, pressing, and loose-ball), recoveries, opponent-half recoveries, possession losses, touches in the box, fouls, and cards. A minimum exposure filter of 600 minutes was then imposed. This reduces the influence of player–season observations whose per-90

or ratio summaries are unstable because they are based on very limited playing time. The filter reduces this instability rather than removing it: Chapter 5 (§2) shows that residual variance still depends on playing time within the filtered sample, which is why the production model scales residual variance by exposure directly rather than relying on the 600-minute threshold alone.

From the aggregated player–season table, the final modelling features were constructed using five pre-processing steps.

DEFINITION 2.1 (Per-90 normalisation). For count variables, per-90 features were computed as

$$x_{ij}^{(90)} = 90 \frac{x_{ij}}{\text{minutes}_i}.$$

This places players with different playing time on a comparable scale and measures action intensity rather than season accumulation.

DEFINITION 2.2 (Efficiency ratios). Where a meaningful denominator was available, success or efficiency variables were computed as

$$r_{ij} = \frac{\text{successful actions}_{ij}}{\text{attempted actions}_{ij}}.$$

Examples include pass accuracy, progressive-pass accuracy, dribble success, defensive-action success, duel-win rate, aerial-win rate, and pressing-duel win rate. These variables separate involvement volume from execution quality.

DEFINITION 2.3 (Log transformation). For a small number of strongly right-skewed non-negative volume variables, a $\log(1 + x)$ transformation was applied:

$$x_{ij}^{(\log)} = \log(1 + x_{ij}).$$

This reduces the influence of extreme observations while preserving the ordering of non-negative values. Five features received this transform: goals, expected goals per shot, expected assists, shot assists, and key passes, all per 90 minutes.

DEFINITION 2.4 (Square-root transformation). A second group of moderately right-skewed volume variables received a milder $\sqrt{x}$ transformation instead:

$$x_{ij}^{(\sqrt{\cdot})} = \sqrt{x_{ij}}.$$

The square root compresses the upper tail less aggressively than $\log(1 + x)$ and was preferred for variables whose skew, while still non-negligible, did not warrant a logarithmic transform. Eighteen features received this transform, spanning most of the passing volume block (total passes, forward, back, lateral, and long passes, received passes, crosses, through passes, progressive passes, smart passes) together with shots, headed shots, shot blocks, recoveries, clearances, interceptions, and fouls suffered, all per 90 minutes. The remaining twenty-five features were left untransformed.

DEFINITION 2.5 (Standardisation). The (possibly transformed) features were standardised across the player–season sample using

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where $\bar{x}_j$ and s_j are the sample mean and standard deviation of feature j on the transformed scale. This prevents variables measured on larger numerical scales from dominating covariance- or distance-based analyses, and is applied after the log/square-root transforms above rather than to the raw counts.

For the exploratory descriptive analyses of Chapter 3, the engineered variables were additionally grouped into five football blocks: offensive production, progression and chance creation, carries and 1v1 actions, defence and pressing, and duels/physical contests (with the small number of discipline-related features folded into the duels/physical block). Equal block weighting was applied so that blocks with more variables did not dominate the representation purely because of dimensionality. Missing values in the final feature matrix were handled at the matrix-construction stage before computing correlations, dissimilarities, and latent representations.

TABLE 1. Summary of the final player–season analytical dataset.

Item	Description
Data provider	Wyscout-derived event data
Storage layer	Professional Redshift warehouse
Competition	LaLiga
Seasons	2011/12 – 2022/23 (12 seasons)
Observational unit	Player–season
Player population	Outfield players only
Minimum exposure	600 minutes per player–season
Number of observations	4586 player–seasons
Number of unique players	1529
Number of active features	48 engineered football features (5 log-transformed, 18 square-root-transformed, 25 untransformed, all Z-scaled)
Metadata retained	Player ID, player name, season ID, competition ID, minutes, matches
Active modelling matrix	Transformed, standardised football features (additionally block-weighted for the exploratory analyses of Chapter 3)

The output of the pipeline is a transformed player–season feature matrix rather than a raw event table. Each row corresponds to one outfield player in one LaLiga season, and each active column corresponds to an engineered football feature after normalisation, transformation, and standardisation. A summary of the final analytical dataset is given in Table 1. This matrix is the input for the dissimilarity calculations and exploratory low-dimensional analyses of Chapter 3, and for the Bayesian hierarchical model of Chapters 4–5.

CHAPTER 3

Background

This chapter introduces the statistical background needed to study similarity between football players from a multivariate latent-space perspective. The main object of interest in this master's thesis is the *player–season observation*, that is, the aggregated statistical profile of a given player over a given season. Each observation is therefore described by multiple performance variables simultaneously, and the central challenge is to understand how such multivariate profiles can be represented, compared, and interpreted in a lower-dimensional space.

More broadly, the aim is to construct statistical representations of player performance that make stylistic similarity empirically assessable. In this setting, similarity is not understood as a directly observed quantity, but as a latent relational structure induced by the joint behaviour of many football variables. This naturally places the problem within the scope of multivariate analysis, where the goal is often to reduce complexity while preserving the main patterns of variation, dependence, and association across observations.

The player–season construction also introduces a repeated-observation component, since some players appear in more than one season. This suggests that player profiles may exhibit persistence or variation over time. This feature of the data is kept in mind as a possible motivation for richer latent-variable formulations if the descriptive and probabilistic analyses indicate that such extensions are warranted.

From a modelling perspective, the chapter develops the background in three parts. First, we review the general multivariate setting of the problem and the idea of representing high-dimensional football data in a smaller number of latent dimensions. Second, we discuss classical low-dimensional descriptive tools—in particular principal component analysis, factor analysis, and classical multidimensional scaling—which are useful for exploratory structure discovery and for identifying interpretable geometric or latent patterns in the data. Third, we introduce probabilistic counterparts to these descriptive ideas, with particular attention to Bayesian latent-space formulations that can incorporate uncertainty directly into the representation.

The need for this background is already visible in the exploratory analysis conducted on the player–season dataset. The data is built from event-based football statistics provided by *Wyscout*, a professional football data platform that collects and structures detailed match event information for scouting and performance analysis. From these event records, season-level aggregates are constructed and organised into the football blocks we defined previously. Once such multivariate profiles are defined, a number of statistical questions arise naturally: how many latent dimensions are needed to summarise the main structure, whether the variables reflect a smaller number of underlying stylistic components, and whether the resulting representation is stable and interpretable enough to support meaningful comparisons between players.

Beyond their descriptive value, these representations may also be relevant for downstream applications in recruitment and squad planning. A low-dimensional latent representation of player–season profiles can provide a structured way to compare footballers, identify stylistic neighbours, and understand how players relate to broader positional or tactical archetypes. In this thesis, several descriptive approaches (principal component analysis, factor analysis, and multidimensional scaling) are explored as candidate tools for this purpose, after which greater emphasis is placed on probabilistic analogues that can represent latent structure while also quantifying uncertainty.

1. Similarity representation

In multivariate analysis, comparing observations requires some rule for translating several variables into a single notion of proximity. This can be done through methods that work directly with covariance or correlation structure, or through a similarity or dissimilarity measure defined on the observed feature vectors. In either case, the underlying issue is the same: different comparison rules emphasize different aspects of the data and may therefore induce different low-dimensional representations.

This point is especially relevant when observations are interpreted as performance profiles. Two player–season observations may be regarded as similar because they have similar magnitudes across variables, because they exhibit similar relative patterns across variables, or because they align along a common latent direction even if their absolute levels differ. Thus, similarity is not a uniquely defined property of the data, but rather a statistical construct that depends on how the multivariate profile is represented and compared.

In the present work, the player–season observations are built from weighted football features summarising different aspects of performance. After suitable preprocessing, standardisation, and block-wise weighting, two comparison rules are considered as initial descriptive choices. The first is the Euclidean distance between observations $x_i, x_j \in \mathbb{R}^p$,

$$d_E(x_i, x_j) = \left(\sum_{k=1}^p (x_{ik} - x_{jk})^2 \right)^{1/2}.$$

This measure is sensitive to overall discrepancies in the feature space and therefore reflects differences in absolute profile location after the chosen transformations have been applied.

The second is a cosine-based dissimilarity, defined as

$$d_C(x_i, x_j) = 1 - \frac{x_i^\top x_j}{\|x_i\| \|x_j\|}.$$

This quantity depends on the angle between the two feature vectors and therefore emphasises profile orientation relative to the origin rather than absolute separation in feature space. Strictly speaking, d_C is not a metric distance: $d_C(x_i, x_j) = 0$ implies that x_i and x_j have the same direction, but not necessarily that $x_i = x_j$.

This distinction is important because this cosine dissimilarity and Pearson correlation use different reference points. Cosine dissimilarity compares vectors relative to the origin, whereas correlation compares centred profiles. In particular, if $\tilde{x}_i = x_i - \bar{x}_i$ and $\tilde{x}_j = x_j - \bar{x}_j$, then

$$\text{corr}(x_i, x_j) = \frac{\tilde{x}_i^\top \tilde{x}_j}{\|\tilde{x}_i\| \|\tilde{x}_j\|}.$$

Thus, correlation is the cosine similarity between centred profiles. This makes correlation relevant for the later PCA and factor-analytic modelling, since these methods are directly based on covariance/correlation structure.

These two measures are not intended to exhaust the set of reasonable comparisons in multivariate statistics. Rather, they provide two conceptually distinct and practically useful benchmarks for the exploratory stage of the analysis: one based more directly on geometric separation in the transformed feature space, and one based on angular separation between multivariate profiles. Their comparison is informative precisely because the resulting similarity structures need not coincide.

The exploratory results in [Figure 1](#) show that Euclidean distance and cosine dissimilarity are substantially related, but not equivalent. This is expected because both are computed from the same transformed feature vectors, while emphasising different aspects of them. Euclidean distance reflects absolute separation in the weighted standardised feature space, whereas cosine dissimilarity reflects angular separation relative to the

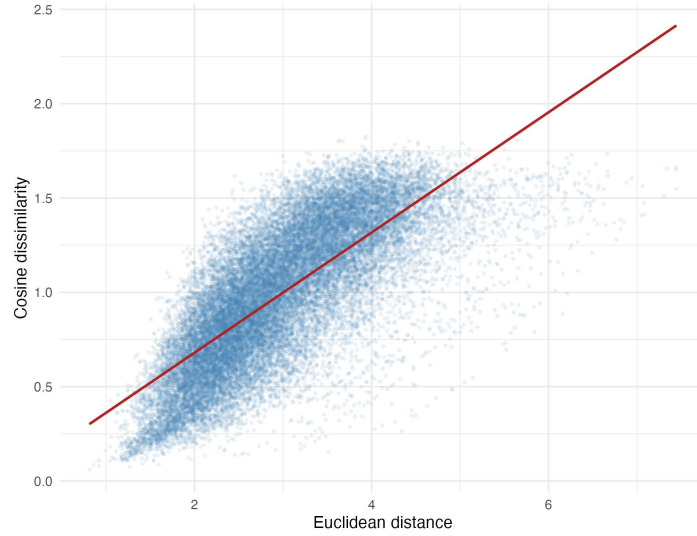


FIGURE 1. Pairwise Euclidean vs. cosine dissimilarities for the player-season subset ($N = 900$, 30,000 sampled pairs). The two metrics show moderate-to-strong agreement (Pearson = 0.746, Spearman = 0.774).

origin. Consequently, the induced neighbourhood structure and the interpretation of player similarity may change depending on the comparison rule.

2. Exploratory low-dimensional structure

A natural question in multivariate analysis is whether a high-dimensional set of observations can be represented satisfactorily in a smaller number of dimensions without losing its main structural patterns. In the present work, this question is approached first through classical exploratory techniques, namely principal component analysis (PCA), factor analysis (FA), and classical multidimensional scaling (CMDS) [14, 3, 4]. These methods do not provide the final probabilistic framework of the work, but they are useful diagnostic tools for determining whether the feature space and the induced distance structure contain stable lower-dimensional patterns worth modelling.

Before specifying a richer latent-variable model, it is helpful to understand whether the data exhibits interpretable low-dimensional structure at all, whether such structure is approximately stable across relevant subsets of it, and whether the dominant structure is more naturally described through covariance/correlation patterns or through dissimilarities. In particular, exploratory analyses can be stratified or inspected by season, playing time, or positional group in order to assess whether a single global representation is plausible or whether the data suggest more heterogeneous structure that should later guide the modelling strategy.

These methods serve different but complementary purposes. PCA works directly on the feature matrix and identifies orthogonal linear combinations that explain as much sample variation as possible [14]. FA also works on the feature matrix, but seeks a lower-rank latent structure capable of reproducing the dependence among the observed variables up to variable-specific residual variation [3]. CMDS, by contrast, starts from a matrix of pairwise dissimilarities and searches for an embedding whose interpoint distances approximate the observed dissimilarities [4]. Taken together, these methods help address several preliminary questions: whether dimension reduction is plausible, whether the dominant axes or factors admit football interpretation, whether the conclusions are robust across relevant subsets of the data, and whether the structure suggested by the features is compatible with the structure suggested by the dissimilarities.

2.1. Principal component analysis. Let $\mathbf{X} = (x_{ij}) \in \mathbb{R}^{n \times p}$ denote the feature matrix, where n is the number of player–season observations and p is the number of transformed and standardised football features retained for the analysis. In the present application, $n = 4586$, corresponding to player–season rows, and the columns of $\mathbf{X}$ represent season-level statistics derived from Wyscout event data, including variables related to shooting, finishing, progression, passing, dribbling, defensive actions, recoveries, duels, and aerial involvement.

Principal component analysis (PCA) is a classical dimension-reduction technique that seeks linear combinations of the observed variables with maximal sample variance under orthogonality constraints [14]. It is useful here as an exploratory device because it reveals whether the player–season feature space is dominated by a small number of broad directions of variation, and because it helps identify which football variables contribute most strongly to those directions.

To define PCA formally, let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ denote the rows of $\mathbf{X}$, and assume that the columns of $\mathbf{X}$ have been centred. For any vector $\mathbf{v} \in \mathbb{R}^p$, define the projected sample scores

$$z_i(\mathbf{v}) = \mathbf{x}_i^\top \mathbf{v}, \quad i = 1, \dots, n,$$

and their sample variance

$$\widehat{\text{Var}}(z(\mathbf{v})) = \frac{1}{n-1} \sum_{i=1}^n (z_i(\mathbf{v}) - \bar{z}(\mathbf{v}))^2.$$

DEFINITION 3.1. The first principal component direction is defined as

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{v}\|=1} \widehat{\text{Var}}(z(\mathbf{v})),$$

and, recursively, for $k \geq 2$, the k -th principal component direction is defined as

$$\mathbf{v}_k = \arg \max_{\substack{\|\mathbf{v}\|=1 \\ \mathbf{v} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1}}} \widehat{\text{Var}}(z(\mathbf{v})).$$

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ are called the principal component directions, and the projected variables $\mathbf{X}\mathbf{v}_1, \dots, \mathbf{X}\mathbf{v}_p$ are called the principal component scores.

A standard result states that these directions are the eigenvectors of the sample covariance matrix of the centred data [14]. If

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X},$$

then the principal component directions are given by the eigenvectors of $\mathbf{S}$, ordered by decreasing eigenvalue. The corresponding eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$ are the sample variances explained by the successive components. If the analysis is instead performed on standardised variables, then one may equivalently work with the sample correlation matrix

$$\mathbf{\Omega} = \frac{1}{n-1} \mathbf{Z}^\top \mathbf{Z}, \quad \mathbf{Z} = \left[\frac{\mathbf{X}_{\cdot 1} - \bar{X}_1 \mathbf{1}}{s_1}, \dots, \frac{\mathbf{X}_{\cdot p} - \bar{X}_p \mathbf{1}}{s_p} \right],$$

where $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation of the j -th variable [14]. The coefficients of each eigenvector are commonly referred to as *loadings*, while the coordinates of the observations after projection onto the component directions are the *scores*.

PCA is used as an exploratory instrument applied only to the transformed football-feature matrix. Season identifiers, competition identifiers, player identifiers, player names, minutes played, and matches are not treated as active PCA variables; they are retained as metadata for interpretation and diagnostic checks. First, PCA provides a rough sense of whether the player–season feature space is intrinsically high-dimensional or whether a moderate number of directions captures a substantial proportion of sample variation. Second,

the leading loadings indicate which football actions most strongly define the dominant axes of variation. Third, the component scores can be inspected across metadata-defined subsets—for instance by season, competition, or minutes played—to assess whether the same low-dimensional structure is broadly visible across the sample.

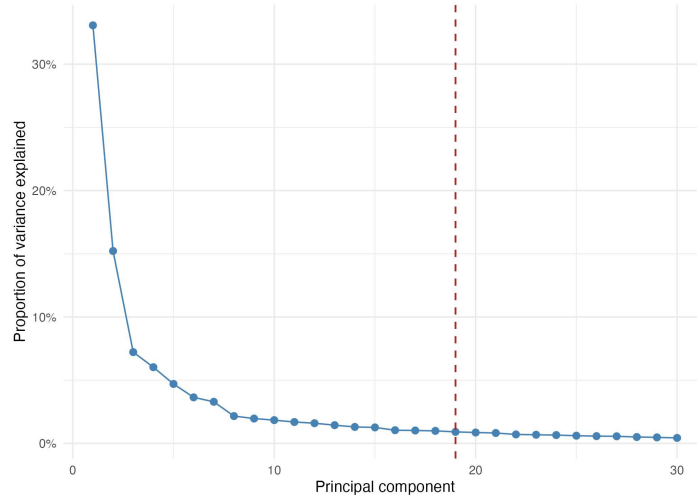


FIGURE 2. Scree plot of the first 30 principal components. PC1 and PC2 jointly explain approximately 48% of the variance, while the first ten components explain approximately 79%. The dashed line marks the first component count reaching at least 90% cumulative explained variance (nineteen components).

The exploratory results suggest that the data are structured but not trivially so. In particular, [Figure 2](#) shows that the first two principal components explain 48.3% of the total sample variance, the first five around 66.2%, and the first ten roughly 79.1%. Using 90% cumulative explained variance as a descriptive threshold requires about nineteen principal components for a high-fidelity linear summary. This indicates that a low-dimensional summary is plausible, but also that the player–season feature space cannot be reduced to a single simple axis without substantial information loss: with $P = 48$ engineered features spanning finishing, passing, carrying, defending, and duelling, considerably more shared structure remains beyond the first few directions than a smaller, coarser feature set would suggest. To interpret the leading components beyond their explained variance, the variables with the largest absolute loadings were inspected. [Table 1](#) reports the five largest absolute loadings for each of the first five principal components.

Interpretations based on the loadings are purely descriptive, and the signs of PCA loadings should not be over-interpreted because eigenvectors are identifiable only up to sign.

Although PCA is informative, its limitations are important. Each component is constructed to be uncorrelated with the previous ones because PCA imposes orthogonality on the extracted directions. This is a mathematical constraint of the method, not evidence that the underlying football traits are independent. For example, ball progression, chance creation, and attacking involvement may be substantively related even if their PCA summaries are forced to lie on orthogonal axes. PCA is therefore interpreted here as a descriptive compression of the transformed feature space, not as a structural model of latent football style.

2.2. Factor analysis. Factor analysis provides an alternative route to low-dimensional structure when the observed variables are thought to reflect a smaller number of underlying latent traits. In contrast to PCA, which is defined through variance-maximising linear projections, factor analysis begins from an explicit statistical model for the covariance structure of the observed variables [3].

TABLE 1. Largest absolute PCA loadings for the first five principal components.

PC1		PC2		PC3	
dribbles	0.318	head shots	-0.298	defensive duels	-0.342
touch in box	0.251	progressive run	0.295	fouls	-0.341
shots	0.244	accelerations	0.279	successful dribbles	0.303
accelerations	0.229	aerial duels	-0.242	loose ball duels	-0.268
progressive run	0.225	dribbles	0.215	dribbles against	-0.263
PC4		PC5			
successful dribbles	0.308	successful dribbles	0.754		
accelerations	-0.299	offensive duels won	0.281		
progressive run	-0.286	accelerations	0.217		
passes to final third	0.228	progressive run	0.177		
through passes	0.223	dribbles	0.171		

DEFINITION 3.2. A factor analysis model for observation i , represented by a random vector $\mathbf{Y}_i \in \mathbb{R}^p$, with k latent factors is given by

$$\mathbf{Y}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda} \mathbf{f}_i + \boldsymbol{\varepsilon}_i,$$

where $\boldsymbol{\mu} \in \mathbb{R}^p$ is a mean vector, $\mathbf{f}_i \in \mathbb{R}^k$ is the latent factor vector for observation i , $\boldsymbol{\Lambda} \in \mathbb{R}^{p \times k}$ is a loading matrix, and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^p$ is an observation-specific residual vector. The matrix $\boldsymbol{\Lambda}$ describes how the observed variables depend linearly on the latent factors, while $\boldsymbol{\varepsilon}$ captures variable-specific residual variation not explained by the common factor structure [3].

Under standard assumptions

$$\mathbb{E}[\mathbf{f}_i] = \mathbf{0}, \quad \text{Cov}(\mathbf{f}_i) = \mathbf{I}_k, \quad \mathbb{E}[\boldsymbol{\varepsilon}_i] = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}_i) = \boldsymbol{\Psi},$$

where $\boldsymbol{\Psi} = \text{diag}(\psi_1, \dots, \psi_p)$, and $\mathbf{f}_i$ is independent of $\boldsymbol{\varepsilon}_i$, it follows that

$$\text{Cov}(\mathbf{Y}) = \boldsymbol{\Lambda} \boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}.$$

Thus, the covariance matrix is decomposed into a low-rank component, $\boldsymbol{\Lambda} \boldsymbol{\Lambda}^\top$, representing shared dependence among the variables, and a diagonal component, $\boldsymbol{\Psi}$, representing variable-specific residual variance.

This framework is appealing in football because the observed features are unlikely to be driven by wholly separate mechanisms. Variables related to progression and creation may share common structure, as may defensive activity and duel intensity, while finishing output may combine contributions from shot volume, box presence, and shot quality. In such a setting, a model that explains dependence through a smaller number of related latent traits is often more plausible than one based only on orthogonal descriptive directions.

Unlike PCA and CMDs, factor analysis is already a probabilistic model. For this thesis, factor analysis provides a simple latent-variable baseline for assessing whether the covariance structure of the player–season feature matrix can be represented through a number of interpretable latent dimensions before specifying a final Bayesian model.

In the exploratory analysis conducted here, after removing a constant feature with zero sample standard deviation, maximum-likelihood exploratory factor analysis was fitted for several candidate numbers of factors, followed by oblique rotation using an oblimin-type criterion in order to allow correlated factors [3, 13].

The number of factors was guided in part by parallel analysis [12]. Parallel analysis compares the observed eigenvalues of the empirical correlation matrix with eigenvalues obtained from random datasets of the same size. Factors are retained while the observed eigenvalues exceed a chosen simulated reference, here the simulated 95th percentile. As shown in Figure 4, the observed eigenvalues dominate the simulated

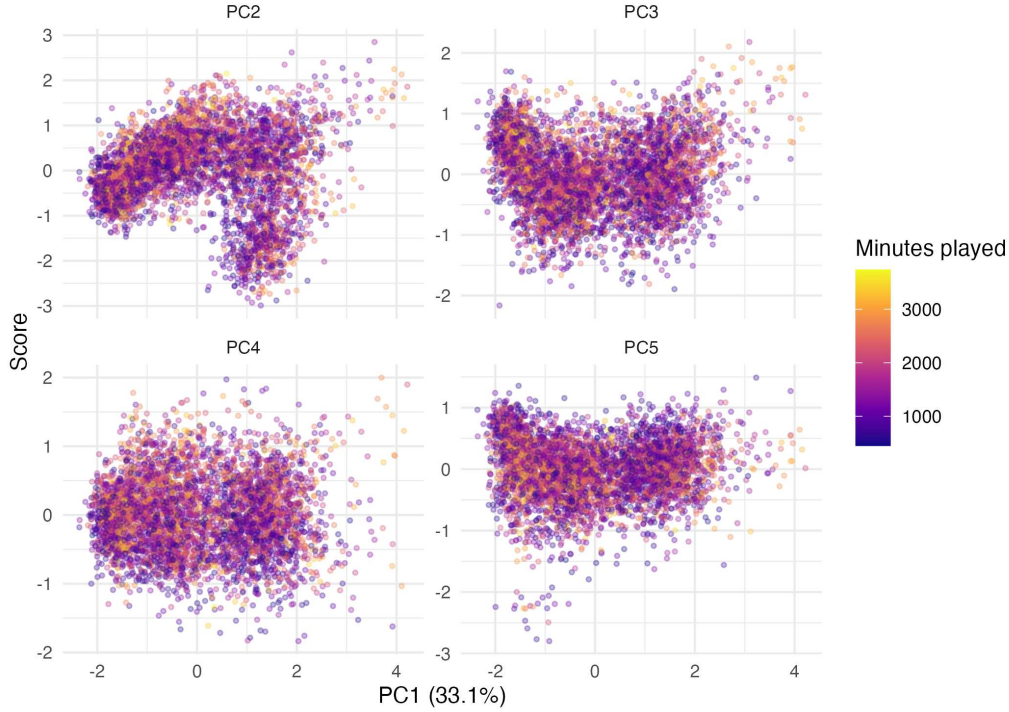


FIGURE 3. Principal component score plots for PC1 against PC2–PC5, with points coloured by minutes played. Playing time does not show a clear monotone gradient along the displayed components, suggesting that minutes played are not the main source of variation in the leading PCA scores.

reference eigenvalues up to the eleventh factor, after which the empirical curve falls below the simulated 95th percentile. This provides empirical support for retaining eleven factors as a reasonable working dimension for the latent covariance structure; the larger count relative to a coarser feature set is expected, since $P = 48$ engineered features resolve considerably more distinct football sub-skills (e.g. separate passing-distance, duel-type, and defensive-action categories) than a smaller feature set would. Competing specifications were then compared using standard fit diagnostics.

More specifically, the root mean square residual (RMSR) summarises the average size of the residual correlations left unexplained by the fitted factor model, while the root mean square error of approximation (RMSEA) measures lack of fit per degree of freedom at the covariance-structure level [3, 5]. The Tucker–Lewis index (TLI), also known as the non-normed fit index, compares the fitted model’s discrepancy per degree of freedom against that of a baseline model in which all variables are uncorrelated, so that values closer to 1 indicate a comparatively better, more parsimony-adjusted fit [26]. Model comparison also considered the Bayesian information criterion (BIC), defined in the sense of Schwarz [21], which balances model fit against complexity.

The exploratory results support the presence of non-trivial latent structure. Parallel analysis suggested an eleven-factor solution, while a nine-factor model provided a more compressed but rougher summary. The eleven-factor specification improved the fit statistics relative to the nine-factor one (TLI rising from 0.804 to 0.843; RMSEA falling from 0.101 to 0.090; RMSR falling from 0.0256 to 0.0227; BIC falling from approximately 28,630 to 19,420), although the overall fit still indicates that the latent structure should be viewed as an approximation rather than a complete explanation of the dependence in the data. Taken

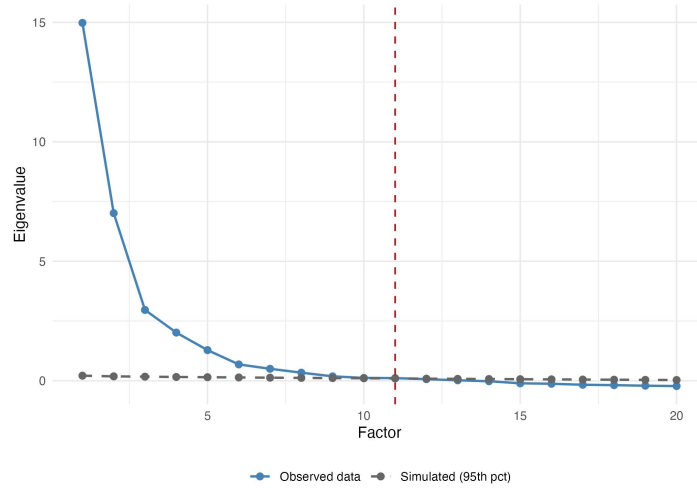


FIGURE 4. Parallel analysis scree plot for factor analysis. The observed eigenvalues are compared with the 95th percentile of eigenvalues obtained from simulated random datasets. The vertical dashed line indicates the suggested number of retained factors, $k = 11$, corresponding to the point after which the observed eigenvalues no longer exceed the simulated reference values.

together, the pattern in Figure 4 and the fit comparisons suggest that eleven factors offer a reasonable compromise between parsimony and interpretability for this larger, $P = 48$ feature set.

Substantively, the rotated loading structure reveals several interpretable latent dimensions; see Figure 5. Ordered by the variance each factor accounts for, the eleven-factor solution separates: chance creation and assists; goal threat and finishing; carrying and offensive duelling; general passing volume and circulation; defensive duel-winning and ball recovery; defensive duel exposure (1v1 defending workload); progressive and vertical passing; physical engagement and possession risk; high/pressing recoveries; passing accuracy versus crossing risk; and deep build-up circulation. While these labels are necessarily interpretive, the loading pattern shows that the latent dimensions are not arbitrary mathematical artefacts, but correspond to recognisable football behaviours – including a clear separation, absent from a coarser feature set, between winning defensive duels and simply being exposed to them, and between general passing volume and specifically progressive, forward-directed passing. Importantly, the estimated factor correlations are not negligible (the strongest, between the chance-creation and carrying/offensive-duelling factors, reaches 0.63, and goal threat correlates negatively with both defensive factors at around -0.5), which reinforces the idea that football style may be better described through related latent traits than through a small set of orthogonal descriptive axes.

The role of factor analysis in this master’s thesis is therefore both exploratory and strategic. It helps assess whether the transformed feature matrix supports an interpretable latent covariance structure, and it provides guidance for the construction of more refined probabilistic models later in the work. At the same time, it remains distinct from approaches based directly on pairwise dissimilarities, so its conclusions should be interpreted as one stage in a sequential modelling strategy rather than as the unique final representation of football similarity.

2.3. Classical multidimensional scaling. Classical multidimensional scaling (CMDS) starts from a matrix of pairwise dissimilarities rather than directly from the feature matrix. When those dissimilarities are ordinary Euclidean distances computed from centred feature vectors, CMDS is closely connected to PCA:

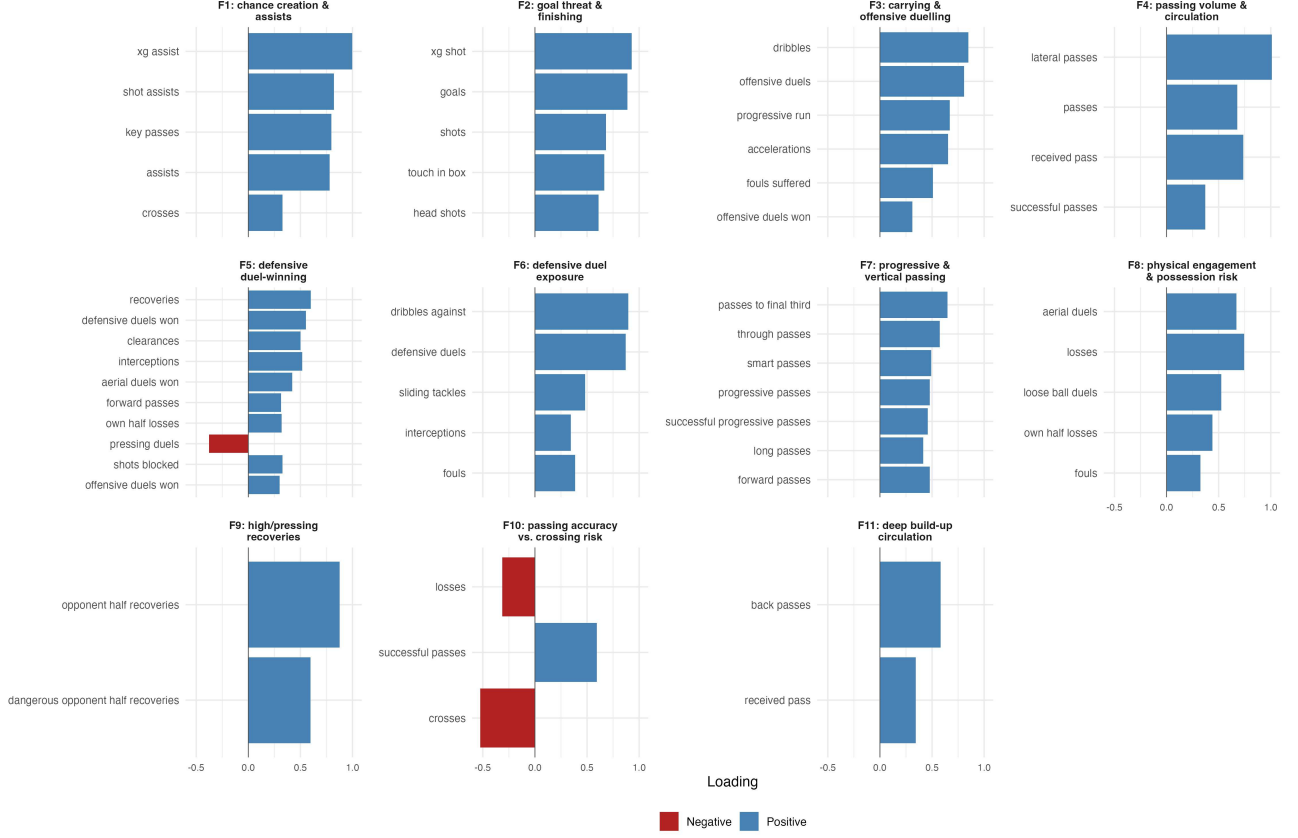


FIGURE 5. Rotated factor loadings for the eleven-factor oblimin solution, displaying variables with $|\text{loading}| \geq 0.30$. Blue and red bars indicate positive and negative loadings, respectively. Panels are ordered by variance explained (F1 highest) and labeled with the corresponding interpretation, matching the enumeration above.

the double-centred squared-distance matrix recovers the same principal-coordinate geometry as the PCA score representation. Its added value here is therefore not to duplicate PCA with Euclidean distances, but to examine non-Euclidean or profile-based dissimilarities, such as cosine-based dissimilarity. This makes it relevant here because one of the exploratory goals is to understand whether the similarity structure induced by the player–season representation can be expressed as a low-dimensional Euclidean configuration [25, 4, 7].

Let $\mathbf{D} = (d_{ij})$ denote the matrix of pairwise dissimilarities between the n observations, with $d_{ii} = 0$, $d_{ij} = d_{ji}$, and let $q < n$ be a target embedding dimension. The CMDS problem is to construct points $\mathbf{z}_1, \dots, \mathbf{z}_n \in \mathbb{R}^q$ such that the Euclidean interpoint distances $\|\mathbf{z}_i - \mathbf{z}_j\|$ approximate the observed dissimilarities as closely as possible. In its classical form, this is achieved through a spectral decomposition of a double-centred matrix of squared dissimilarities [25, 4].

THEOREM 3.3. Suppose that $\mathbf{D}^{(2)} = (d_{ij}^2)$ is the matrix of squared pairwise Euclidean distances generated by some centred configuration of n points in $\mathbb{R}^m$. Let

$$\mathbf{J} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

be the centring matrix, and define

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J}.$$

Then $\mathbf{B}$ is the Gram matrix of the centred configuration. If

$$\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$$

is its spectral decomposition, then the coordinates of the configuration can be recovered, up to rigid transformations, as

$$\mathbf{Z} = \mathbf{V}\mathbf{\Lambda}^{1/2}.$$

If only the first q positive eigenvalues are retained, so that

$$\mathbf{Z}_q = \mathbf{V}_q\mathbf{\Lambda}_q^{1/2},$$

then $\mathbf{Z}_q$ provides the best rank- q approximation to $\mathbf{B}$ in the least-squares sense given by the Eckart–Young theorem, and hence the classical q -dimensional CMDS representation obtained through spectral truncation [25, 7, 4].

Thus, the double-centred matrix provides a closed-form route to principal coordinates when the input dissimilarities are squared Euclidean distances. In that case, CMDS is essentially a distance-based route to the same geometry obtained from PCA on the centred data. When the input dissimilarities are not squared Euclidean, as with cosine-based dissimilarity in the present exploratory analysis, the same spectral procedure gives an approximate Euclidean representation of that dissimilarity structure [7, 4]. In practice, the eigenvalue pattern of $\mathbf{B}$ is therefore informative about whether the distance structure is close to low-dimensional Euclidean and about how much information is lost when only a few dimensions are retained.

In the present work, CMDS was applied in particular to the cosine-based dissimilarity matrix. This matrix should be understood as an angular-based dissimilarity matrix rather than as a strict metric distance matrix. Cosine-based dissimilarity emphasizes the direction of the player profile rather than its overall norm, and when vectors are centred this directional comparison is equivalent to correlation. In this sense, two player–season observations may be regarded as similar not because they have the same absolute output, but because they exhibit a similar pattern across features. For example, two players with very different total event volumes may still be close under cosine-based dissimilarity if both display the same relative emphasis on progression, dribbling, defensive work, and duel involvement. CMDS then provides a first low-dimensional map of this pattern-based similarity structure.

This representation is useful for exploratory purposes. It allows one to inspect whether the player–season observations occupy a coherent geometric space, whether certain profiles appear as extremes of the representation, and whether metadata-defined subsets of the data—for instance observations from different seasons, positional groups, or usage levels—occupy comparable regions of the map. At the same time, the limitations of CMDS are clear. It yields a deterministic embedding rather than a probabilistic one, provides no direct quantification of uncertainty for player locations, and does not by itself account for repeated observations of the same player across seasons or for more structured hierarchical variation.

The labelled extremes in Figure 6b help interpret the two displayed dimensions. The positive side of Dimension 1 is anchored mainly by wide attacking profiles, such as F. Orellana, Mikel Oyarzabal, Iñaki Williams, and I. Cavaleiro, whereas the negative side contains centre-back profiles, such as G. Cabral, Carlos Domínguez, David López, David Costas, and S. Savić. Dimension 2 separates aerial or penalty-box reference forwards, such as A. Véliz, S. Rondón, Hugo Duro, and Y. En-Nesyri, from attacking full-back profiles with substantial progression involvement, such as Dani Carvajal, Filipe Luís, M. Acuña, and Sergi Roberto. These labels should be interpreted descriptively: they identify the observed profiles anchoring the CMDS axes, not fixed universal football roles. For these reasons, CMDS is best viewed here as an exploratory geometric benchmark rather than as the final model. It indicates whether the chosen dissimilarity admits a coherent

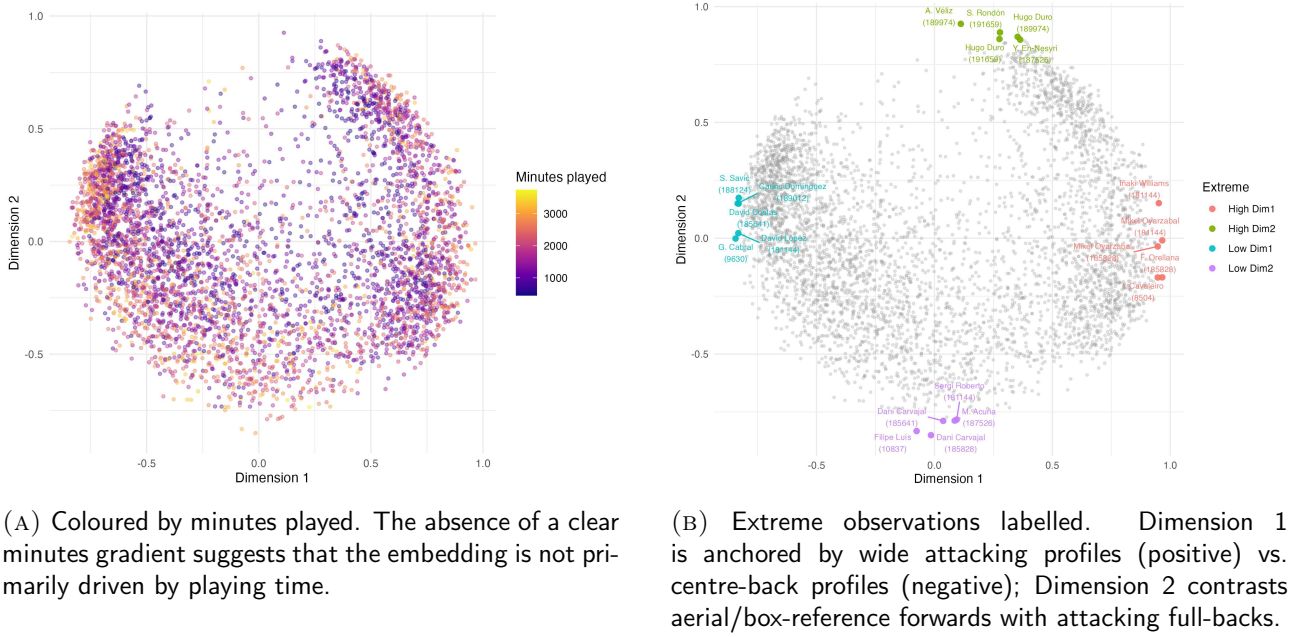


FIGURE 6. Two-dimensional CMDS embedding of the player–season observations using cosine-based dissimilarity.

low-dimensional approximation, and it provides a descriptive precursor to more structured probabilistic latent-space models, including Bayesian and hierarchical extensions [18, 28].

2.4. Summary of the exploratory role of PCA, FA, and CMDS. Taken together, PCA, FA, and CMDS provide complementary descriptive evidence that the player–season data contain meaningful lower-dimensional structure. PCA suggests that a moderate number of directions captures a large proportion of total variation. FA indicates that several correlated latent traits may underlie the observed features. CMDS shows that the pairwise dissimilarity structure admits an interpretable low-dimensional Euclidean approximation.

These findings do not by themselves determine the final model, but they justify moving towards a latent-space formulation. More importantly, they reveal that the data contain enough structure for such a model to be meaningful, while also making clear that a purely descriptive approach is insufficient if one wishes to quantify uncertainty and account for the longitudinal character of player–season observations.

CHAPTER 4

Model

1. Motivation and overview

The exploratory analyses presented in Chapter 3 establish that the player–season feature space contains meaningful lower-dimensional structure. Principal component analysis revealed that approximately nineteen components are needed to account for ninety per cent of the total sample variance, while an eleven-factor oblimin solution provided an interpretable account of the empirical correlation structure through recognisable football dimensions such as goal threat, chance creation, pass circulation, defensive work, and duel efficiency. Classical multidimensional scaling applied to the cosine-based dissimilarity confirmed that the pairwise similarity structure admits a low-dimensional Euclidean approximation. Taken together, these findings support the view that player–season performance profiles are governed by a moderate number of shared latent traits, and that the feature space is rich enough for such traits to be recoverable from the data.

What the exploratory analyses cannot do, however, is quantify uncertainty in the latent structure. PCA scores and factor loadings are point estimates computed from the sample; they carry no natural notion of sampling variability, posterior distribution, or credible interval. This matters especially in a longitudinal setting, where the same player is observed in multiple seasons. The exploratory analyses treat all player–season rows as exchangeable, whereas a proper account of the data-generating process recognises that two observations belonging to the same player are not independent: they share the player’s stable latent characteristics. Ignoring this structure risks confusing player persistence with seasonal noise, and overstating the precision of latent-dimension estimates.

This chapter specifies a probabilistic model designed to address these limitations. The model is hierarchical: each player’s observed performance vector in a given season is the sum of three parts; a stable player effect, a season-specific shift, and residual noise. The player effects are themselves modelled as draws from a distribution whose covariance is structured as a low-rank factor model, directly encoding the finding from Chapter 3 that player-level variation is largely explained by a small number of shared latent dimensions.

The model has three principal objectives. First, it provides a principled variance decomposition, attributing each feature’s total variance to player, season, and residual components; this decomposition is expressed as a set of intraclass correlation coefficients, one per feature. Second, it estimates the covariance structure of the player effects through a factor model with K shared latent dimensions, recovering interpretable directions in the player-effect space. Third, all inferences are Bayesian: the model produces a joint posterior distribution over all parameters, from which uncertainty in the latent structure, the variance decomposition, and player-level factor scores is directly quantifiable [6].

2. Notation

Throughout this chapter, the following notation is used consistently. Let N denote the total number of player–season observations, I the number of unique players, S the number of seasons, P the number of standardised performance features, and K a positive integer referred to as the factor rank, whose selection will be discussed in Section 8. Observation indices run from $n = 1, \dots, N$; $i[n] \in \{1, \dots, I\}$ and $j[n] \in \{1, \dots, S\}$ are the player and season indices of observation n , respectively.

TABLE 1. Summary of notation used throughout this chapter.

Symbol	Meaning
N, I, S, P, K	Total observations; unique players; seasons; features; factor rank
$n = 1, \dots, N$	Observation index
$i[n] \in \{1, \dots, I\}$	Player index of observation n
$j[n] \in \{1, \dots, S\}$	Season index of observation n
$\mathbf{y}_n \in \mathbb{R}^P$	Standardised performance vector for observation n
$\mathbf{a}_i \in \mathbb{R}^P$	Player effect for player i
$\mathbf{b}_j \in \mathbb{R}^P$	Season effect for season j
$\varepsilon_n \in \mathbb{R}^P$	Residual for observation n
$\mathbf{\Lambda} \in \mathbb{R}^{P \times K}$	Factor loading matrix
$\mathbf{\Psi}_a = \text{diag}(\psi_{a,1}^2, \dots, \psi_{a,P}^2)$	Uniqueness matrix
$\mathbf{\Sigma}_a, \mathbf{\Sigma}_b, \mathbf{\Sigma}_e$	Player, season, and residual covariance matrices
$\boldsymbol{\eta}_i \in \mathbb{R}^K$	Latent factor scores for player i
$\mathbf{z}_{u,i} \in \mathbb{R}^P$	Latent uniqueness scores for player i
$\nu > 2$	Degrees-of-freedom parameter of the Student- t residual
$\phi \geq 0$	Observation-level variance scaling exponent
m_n	Minutes played in observation n
m_{ref}	Reference exposure level

3. The additive hierarchical decomposition

The central structural assumption of the model is that the performance vector for any given player–season observation arises from an additive decomposition into three independent latent components.

DEFINITION 4.1 (Additive decomposition). For each observation $n = 1, \dots, N$, the observed performance vector $\mathbf{y}_n \in \mathbb{R}^P$ satisfies

$$(1) \quad \mathbf{y}_n = \mathbf{a}_{i[n]} + \mathbf{b}_{j[n]} + \varepsilon_n,$$

where $\mathbf{a}_i \in \mathbb{R}^P$ is the *player effect* for player i , $\mathbf{b}_j \in \mathbb{R}^P$ is the *season effect* for season j , and $\varepsilon_n \in \mathbb{R}^P$ is the *residual* for observation n . The three components are assumed mutually independent.

The player effect $\mathbf{a}_i$ captures those aspects of player performance that are stable across seasons: the passing volume, defensive involvement, and goal-threat profile that characterise a player’s identity rather than a particular team context. The season effect $\mathbf{b}_j$ captures shifts in the distribution of performance statistics that are common to all players in season j : changes in the pace of play, officiating style, or data-collection conventions that may change the observed values of many features simultaneously. The residual ε_n captures what remains after both systematic components are removed, including player–season idiosyncrasy, noise from the finite number of matches played, and any tactical or physical changes specific to one season that are not representative of the player’s profile.

REMARK 4.2 (Absence of a global mean). No explicit population mean vector $\boldsymbol{\mu}$ appears in (1). This is because the features $\mathbf{y}_n$ are constructed by standardising each feature to zero mean and unit variance across the full sample, as described in Chapter 2. The population mean is therefore absorbed into the standardisation step, and the model is centred from the outset.

The decomposition (1) defines a *crossed* random-effects structure: players and seasons are crossed, not nested, each player appears in several seasons, and each season contains many players. This is in contrast

to a nested structure in which seasons would be defined within players. The crossed structure is appropriate here because the inferential goal is to separate player-level persistence from season-level context, not to model players as a random sample within fixed seasons [9].

4. Covariance structure

The three components $\mathbf{a}_i$, $\mathbf{b}_j$, and ε_n are each assigned a zero-mean multivariate distribution; they differ only in the structure of their covariance matrices. The choices here for said components differ: a richer covariance structure is assigned to the player effects, which are the primary object of interest and for which the data contain many observations per latent quantity, while simpler structures are assigned to season effects and residuals.

4.1. Player covariance: low-rank plus diagonal.

DEFINITION 4.3 (Factor covariance for player effects). The player effects $\mathbf{a}_i$ are exchangeable with zero mean and covariance

$$(2) \quad \Sigma_a = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \Psi_a,$$

where $\mathbf{\Lambda} \in \mathbb{R}^{P \times K}$ is the loading matrix and $\Psi_a = \text{diag}(\psi_{a,1}^2, \dots, \psi_{a,p}^2)$ is the uniqueness matrix.

The structure in (2) decomposes the player-effect covariance into two parts. The term $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ is a rank- K positive semi-definite matrix that captures the shared dependence among features through K latent dimensions; Ψ_a is a diagonal matrix that captures the feature-specific variance not explained by those shared dimensions.

PROPOSITION 4.4 (Entries of Σ_a). The diagonal entry $[\Sigma_a]_{pp}$ for feature p is

$$(3) \quad [\Sigma_a]_{pp} = \sum_{k=1}^K \Lambda_{pk}^2 + \psi_{a,p}^2,$$

and the off-diagonal entry for features $p \neq q$ is

$$(4) \quad [\Sigma_a]_{pq} = \sum_{k=1}^K \Lambda_{pk} \Lambda_{qk}.$$

PROOF. Both statements follow by expanding $[\mathbf{\Lambda}\mathbf{\Lambda}^\top + \Psi_a]_{pq}$. For the diagonal, $[\mathbf{\Lambda}\mathbf{\Lambda}^\top]_{pp} = \sum_k \Lambda_{pk}^2$ and $[\Psi_a]_{pp} = \psi_{a,p}^2$. For the off-diagonal, $[\mathbf{\Lambda}\mathbf{\Lambda}^\top]_{pq} = \sum_k \Lambda_{pk} \Lambda_{qk}$ and $[\Psi_a]_{pq} = 0$ since Ψ_a is diagonal. $\square$

In the factor-analysis literature, the quantity $\sum_{k=1}^K \Lambda_{pk}^2$ is called the *communality* of feature p : it measures how much of the player-level variance in feature p is explained by the K shared latent dimensions. The quantity $\psi_{a,p}^2$ is called the *uniqueness*: it is the portion of player-level variance in feature p that is specific to that feature and not captured by the shared structure. Together, they account for all player-level variance in feature p , as stated in (3).

The off-diagonal entries in (4) reflect the pairwise covariance between features p and q induced by the shared factors: two features co-vary at the player level only to the extent that they share common loadings on the K dimensions. A player who scores highly on a latent factor tends to score highly on all features with positive loadings on that factor, creating positive covariance among those features. The uniqueness matrix Ψ_a contributes nothing to off-diagonal entries, reflecting the assumption that, conditional on the K shared factors, feature-specific residuals are uncorrelated.

This covariance structure is motivated directly by the exploratory findings of Chapter 3. There, classical factor analysis showed that the 48 features are not independent at the player level: they cluster into correlated

groups, and a number of shared latent traits (eleven, by parallel analysis) were needed to reproduce their correlation pattern.

The low-rank form $\Sigma_a = \Lambda\Lambda^\top + \Psi_a$ builds that same idea into the model's assumptions, rather than just describing it after the fact. It says exactly this: player-level covariance among features is mostly explained by K shared dimensions ($\Lambda\Lambda^\top$), plus a small, purely feature-specific leftover (Ψ_a).

The difference from the earlier exploratory chapter is that this is now a fully probabilistic (Bayesian) model, not a one-off descriptive result [17]. Classical factor analysis gives a single point-estimate set of loadings, with no sense of how uncertain they are. Here, Λ , Ψ_a , and everything built from them instead get a full posterior distribution: the model does not just claim that shared structure exists, it quantifies how confident it is about it. The rank K actually adopted for Σ_a is selected separately in Section 8, and Remark 4.19 explains why it need not, and does not, match the raw factor count above.

4.2. Season covariance: diagonal.

REMARK 4.5 (Diagonal season covariance). The season effects $\mathbf{b}_j$ are assigned zero mean and diagonal covariance:

$$(5) \quad \Sigma_b = \text{diag}(\sigma_{b,1}^2, \dots, \sigma_{b,P}^2).$$

This is an asymmetric design choice relative to the player covariance. A richer specification, $\Sigma_b = \Lambda_b\Lambda_b^\top + \Psi_b$, would model correlated season-level drift, but would require reliable estimation of $P \times K_b$ loading parameters from only S observations of the season. With $S = 12$ seasons, this is severely under-determined for any $K_b \geq 1$: preliminary fits with a low-rank Σ_b produced R-hat values exceeding 1.70 and bulk effective sample sizes as low as 6 for the season loadings, indicating that the posterior was entirely dominated by the prior. The diagonal specification (5) allows each feature its own season-level drift magnitude while avoiding parameters that the data cannot identify, plus the computational convenience.

4.3. Residual covariance. The residuals ε_n are assigned a diagonal covariance $\Sigma_e = \text{diag}(\sigma_{e,1}^2, \dots, \sigma_{e,P}^2)$, reflecting the assumption that, conditional on the player and season effects, the P features are uncorrelated at the observation level. The diagonal entries themselves are not constant across observations in the production model: Section 5.3 makes $\sigma_{e,p}$ depend on the observation's playing time, $\sigma_{e,n,p}$, to account for the fact that per-90 features are noisier estimates for players with fewer minutes. The cross-feature independence encoded by the diagonal structure is unaffected by that refinement.

5. Likelihood specification

The additive decomposition (1) and the covariance structure of Section 4 are common to all model variants considered here. The models differ in the distributional form assigned to the scalar residuals ε_{np} , the p -th component of ε_n .

5.1. Normal baseline. The simplest complete specification assigns each residual component a Gaussian distribution:

$$(6) \quad \varepsilon_{np} \mid \sigma_{e,p} \sim \mathcal{N}(0, \sigma_{e,p}^2), \quad p = 1, \dots, P,$$

independently across features and observations, giving the conditional joint likelihood

$$(7) \quad \mathbf{y}_n \mid \mathbf{a}_{i[n]}, \mathbf{b}_{j[n]}, \sigma_e \sim \mathcal{N}_P(\mathbf{a}_{i[n]} + \mathbf{b}_{j[n]}, \Sigma_e).$$

This specification, referred to as the Normal model, serves as a computational and inferential baseline against which richer likelihood choices are compared.

5.2. Student- t residuals. For player–season data, the Gaussian assumption on residuals is restrictive. Some player–seasons produce extreme per-90 statistics for identifiable reasons: a player returning from injury may appear in only a handful of matches with atypical physical load; a player deployed in an unusual tactical role for one season may display statistics far from any stable average; a breakout season may represent an extreme but transient realisation. Under the Normal model, these extreme residuals exert a disproportionate influence on the estimates of the player and season effects, since the Gaussian density penalises departures from the conditional mean quadratically and without limit. A heavier-tailed distribution assigns such observations a higher residual likelihood and effectively down-weights their influence on the shared parameters.

DEFINITION 4.6 (Student- t error model). In the Student- t model, the residual for feature p in observation n follows

$$(8) \quad \varepsilon_{np} \mid \nu, \sigma_{e,p} \sim t(\nu, 0, \sigma_{e,p}), \quad p = 1, \dots, P,$$

independently across features and observations, where $\nu > 2$ is a degrees-of-freedom parameter shared across all features and observations, and $\sigma_{e,p} > 0$ is the feature-specific scale. The Student- t distribution with ν degrees of freedom, location zero, and scale σ has density

$$(9) \quad f(\varepsilon; \nu, \sigma) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi} \sigma} \left(1 + \frac{\varepsilon^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}.$$

PROPOSITION 4.7 (Variance of the Student- t residual). *Under Definition 4.6, the variance of ε_{np} is*

$$(10) \quad \text{Var}(\varepsilon_{np}) = \sigma_{e,p}^2 \frac{\nu}{\nu - 2},$$

which is finite for $\nu > 2$ and converges to $\sigma_{e,p}^2$ as $\nu \rightarrow \infty$.

PROOF. The variance of a Student- t random variable with ν degrees of freedom and scale σ is $\sigma^2\nu/(\nu - 2)$, valid for $\nu > 2$. As $\nu \rightarrow \infty$, $\nu/(\nu - 2) \rightarrow 1$. $\square$

The tail-weight parameter ν has a direct interpretation: small values of ν produce heavier tails and greater likelihood of extreme residuals; large values recover the Gaussian behaviour [16, 10]. The degrees-of-freedom parameter is treated as unknown and estimated from the data; its posterior summarises the evidence in favour of heavy-tailed residuals conditional on the model specification and the prior.

Proposition 4.7 assumes the constant residual scale $\sigma_{e,p}$ of this section's base specification. Section 5.3 replaces $\sigma_{e,p}$ by an observation-dependent scale in the production model, and the residual variance below is restated accordingly (Corollary 4.9).

5.3. Observation-level variance scaling. The residual scale $\sigma_{e,p}$ in (6) and (8) is constant across observations. Section 5.2's Student- t specification is stated this way because it is the natural point to introduce heavy tails on their own, but a constant residual scale does not hold for this data and is not what the production model uses. The performance features $\mathbf{y}_n$ are constructed as per-90 rates or efficiency ratios, both of which are ratio estimators: for a player who accumulated m_n minutes, the sampling variance of the per-90 estimate of feature p is approximately proportional to m_n^{-1} , independently of any structural player or season effects. An observation with 600 minutes of playing time is therefore inherently noisier than one with 2,500 minutes, even if both belong to players with identical latent profiles.

Left unaddressed, this heteroscedasticity causes the model to attribute the elevated variance of low-minute observations to heavy residual tails (via a smaller ν) instead of to exposure, which is a misspecification, not a modelling nicety to defer. The production model removes it by scaling the residual variance at the observation level, as follows.

DEFINITION 4.8 (Minutes-weighted residual scale). Let $m_{\text{ref}} > 0$ denote a reference exposure level, taken as the sample median of the minutes-played values $\{m_n\}_{n=1}^N$. Define the observation-level residual standard deviation for observation n and feature p as

$$(11) \quad \sigma_{e,n,p} = \sigma_{e,p} \cdot \left(\frac{m_{\text{ref}}}{m_n} \right)^\phi,$$

where $\phi \geq 0$ is the scaling exponent. The Student- t model with minutes-weighting replaces $\sigma_{e,p}$ in (8) with $\sigma_{e,n,p}$. This is the residual specification used by the production model throughout the remainder of this chapter and in Chapter 5: Definition 4.6 combined with (11), not Definition 4.6 in isolation.

COROLLARY 4.9 (Variance under the minutes-weighted scale). Replacing $\sigma_{e,p}$ by $\sigma_{e,n,p}$ in Proposition 4.7 gives

$$(12) \quad \text{Var}(\varepsilon_{np}) = \sigma_{e,n,p}^2 \frac{\nu}{\nu - 2} = \sigma_{e,p}^2 \left(\frac{m_{\text{ref}}}{m_n} \right)^{2\phi} \frac{\nu}{\nu - 2},$$

which depends on the observation n through m_n and reduces to Proposition 4.7's expression when $m_n = m_{\text{ref}}$.

Three canonical values of ϕ are immediately interpretable. Setting $\phi = 0$ recovers the constant-scale case of (8). Setting $\phi = 1/2$ corresponds to a square-root scaling analogous to the standard deviation of a Poisson count: variance grows linearly with event rate and standard deviation grows as the square root of inverse minutes. Setting $\phi = 1$ imposes full inverse-minutes scaling, so that the residual standard deviation is proportional to $m_n^{-1/2}$, as it would be for a simple average estimator under a constant-intensity Poisson process.

REMARK 4.10 (ϕ is fixed, not estimated). The production model fixes $\phi = 1/2$, the Poisson-rate value described above, rather than treating it as a free parameter. Fixing ϕ introduces no additional parameter and requires no prior on ϕ in (18). The alternative — assigning ϕ a prior and sampling it jointly with the remaining parameters — is a well-posed model: identifiability of a free ϕ rests on the contrast between the residuals of high- and low-minute observations, since, given estimates of $\mathbf{a}_i$ and $\mathbf{b}_j$, the relationship between $|\varepsilon_{np}|$ and m_n is informative about ϕ , though ϕ must be identified jointly with $\sigma_{e,p}$ and ν , all three of which govern the scale and shape of the marginal residual distribution. This alternative was fit as a diagnostic check, with ϕ assigned a Normal(0.5, 0.2) prior (Chapter 5): its posterior concentrates below 0.5 but close to it, which is read there as support for fixing ϕ at the theoretically motivated value rather than as a reason to prefer the estimated one.

6. Model identification

The factor model (2) introduces a structural non-identifiability in the loading matrix $\mathbf{A}$ that must be resolved before Bayesian inference is defined.

6.1. Rotational non-identifiability.

PROPOSITION 4.11 (Rotation equivalence). For any orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{K \times K}$ satisfying $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_K$, the loading matrices $\mathbf{A}$ and $\mathbf{A}\mathbf{Q}$ generate the same covariance:

$$(13) \quad (\mathbf{A}\mathbf{Q})(\mathbf{A}\mathbf{Q})^\top = \mathbf{A}\mathbf{Q}\mathbf{Q}^\top\mathbf{A}^\top = \mathbf{A}\mathbf{A}^\top,$$

and consequently $(\mathbf{A}\mathbf{Q})(\mathbf{A}\mathbf{Q})^\top + \boldsymbol{\Psi}_a = \boldsymbol{\Sigma}_a$.

PROOF. Direct expansion using $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_K$. □

Proposition 4.11 implies that the map $(\mathbf{\Lambda}, \Psi_a) \mapsto \Sigma_a$ is many-to-one: any element of the orbit $\{\mathbf{\Lambda}\mathbf{Q} : \mathbf{Q} \in O(K)\}$ produces the same covariance, where $O(K)$ denotes the group of $K \times K$ orthogonal matrices. The group $O(K)$ has dimension $K(K-1)/2$, corresponding to the degrees of freedom of a K -dimensional rotation; it also includes reflections, so that there are 2^K discrete sign symmetries in addition to the continuous rotational freedom. Without constraints, the posterior distribution over $\mathbf{\Lambda}$ is multimodal and the sampler cannot mix across symmetry-equivalent modes.

6.2. Lower-triangular positive-diagonal parametrisation. A classical and computationally convenient solution to the non-identifiability described above is to impose a constraint on $\mathbf{\Lambda}$ that selects exactly one representative from each equivalence class under $O(K)$.

DEFINITION 4.12 (LT-PD constraint). A matrix $\mathbf{\Lambda} \in \mathbb{R}^{P \times K}$ satisfies the *lower-triangular positive-diagonal* (LT-PD) constraint if:

- (i) $\Lambda_{pk} = 0$ for all $p < k$ (lower-triangular structure), and
- (ii) $\Lambda_{kk} > 0$ for all $k = 1, \dots, K$ (positive diagonal).

PROPOSITION 4.13 (Identification via LT-PD). Suppose $P \geq K$ and let $\mathcal{L}$ denote the set of $P \times K$ matrices satisfying the LT-PD constraint. Then the map

$$\mathcal{L} \times \{\Psi_a : \psi_{a,p} > 0 \forall p\} \longrightarrow S_P^+, \quad (\mathbf{\Lambda}, \Psi_a) \mapsto \mathbf{\Lambda}\mathbf{\Lambda}^\top + \Psi_a,$$

is injective, where S_P^+ denotes the cone of $P \times P$ positive definite matrices. The factor covariance model is therefore identified under the LT-PD constraint.

The result in Proposition 4.13 is established in [1]; the lower-triangular structure eliminates the continuous rotational freedom by zeroing out the $K(K-1)/2$ upper-triangular entries, while the positive-diagonal constraint eliminates the 2^K discrete sign symmetries. An accessible treatment in the Bayesian factor analysis context is given in [11].

REMARK 4.14 (The LT-PD constraint is not an interpretive rotation). The LT-PD constraint resolves the mathematical non-identifiability [1, 11], but it does not produce loadings that are interpretable in a football sense. A simple-structure rotation such as varimax or oblimin minimises cross-loadings and maximises the separation of features across factors, yielding loading patterns that correspond to recognisable dimensions. The LT-PD constraint does neither: it fixes a coordinate system by zeroing out the upper triangle of $\mathbf{\Lambda}$, a choice determined by feature ordering rather than by football logic. The resulting columns of $\mathbf{\Lambda}$ do not correspond to canonical latent directions and should not be reported or interpreted as such. Interpretable factor directions are recovered through the post-processing procedure described in Section 10.

7. Prior specification

All model parameters are assigned priors that are weakly informative relative to the scale of the Z-standardised outcome $\mathbf{y}_n$. Because the features have unit variance by construction, a loading of magnitude $|\Lambda_{pk}| = 1$ represents a one-standard-deviation shift in feature p per unit change in the k -th factor score, which is already a large effect for a Z-scored performance variable. Priors concentrated near zero with tails extending to values of order one are therefore practically and scientifically reasonable.

DEFINITION 4.15 (Priors on the loading matrix). The diagonal entries of $\mathbf{\Lambda}$ are assigned

$$(14) \quad \Lambda_{kk} \sim \text{Half-Normal}(0, 1), \quad k = 1, \dots, K,$$

consistent with the LT-PD positivity constraint. The free lower-triangular entries are assigned

$$(15) \quad \Lambda_{pk} \sim \mathcal{N}(0, 1), \quad 1 \leq k < p \leq P.$$

DEFINITION 4.16 (Priors on variance parameters). The uniqueness standard deviations, the season standard deviations, and the residual standard deviations are each assigned independent Half-Normal priors:

$$(16) \quad \begin{aligned} \psi_{a,p} &\sim \text{Half-Normal}(0, 1), \\ \sigma_{b,p} &\sim \text{Half-Normal}(0, 1), \\ \sigma_{e,p} &\sim \text{Half-Normal}(0, 1), \quad p = 1, \dots, P. \end{aligned}$$

The Half-Normal(0, 1) prior for scale parameters concentrates most mass near zero while permitting values of order one with positive probability; it is recommended for variance components in hierarchical models as a weakly informative alternative to flat or improper priors, which can produce improper posteriors in hierarchical settings [8].

DEFINITION 4.17 (Prior on the degrees-of-freedom parameter). The Student- t degrees-of-freedom parameter is assigned

$$(17) \quad \nu \sim \text{Gamma}(2, 0.1),$$

supported on $\nu > 2$ to ensure finite residual variance. The Gamma(2, 0.1) distribution has mean 20 and standard deviation approximately 14. It places substantial mass below $\nu = 10$, heavy tails, while also allowing the posterior to concentrate near larger values if the data support behaviour close to Gaussian. The prior does not favour either the Gaussian limit $\nu \rightarrow \infty$ nor the extreme heavy-tail regime $\nu \rightarrow 2^+$ [15].

Since the production model fixes $\phi = 1/2$ (Remark 4.10) rather than estimating it, ϕ is not a random variable and is not assigned a prior here; it enters (11) as a known constant. (The diagnostic fit mentioned in Remark 4.10, which does estimate ϕ , assigns it $\phi \sim \text{Normal}(0.5, 0.2)$ — reported there for completeness, not as part of the production prior stack below.)

The joint prior is the product of all terms in Definitions 4.15–4.17, since all parameters are treated as independent then the joint prior density is:

$$(18) \quad \begin{aligned} p(\mathbf{\Lambda}, \psi_a, \sigma_b, \sigma_e, \nu) &= \prod_{k=1}^K p_{\text{HN}}(\Lambda_{kk}) \prod_{\substack{p=1 \\ p > k}}^P \mathcal{N}(\Lambda_{pk}; 0, 1) \\ &\times \prod_{p=1}^P p_{\text{HN}}(\psi_{a,p}) p_{\text{HN}}(\sigma_{b,p}) p_{\text{HN}}(\sigma_{e,p}) \\ &\times p_{\text{Ga}}(\nu), \end{aligned}$$

where p_{HN} denotes the Half-Normal(0, 1) density and p_{Ga} denotes the Gamma(2, 0.1) density; ϕ does not appear, being fixed rather than estimated.

8. Rank selection

The factor rank K governs the dimension of the latent space used to model the player covariance $\mathbf{\Sigma}_a$. Unlike the parameters $\mathbf{\Lambda}$, $\mathbf{\Psi}_a$, and ν , the rank K is not a continuous model parameter that can be assigned a prior and sampled within a single model: it determines the parametric family itself. The selection of K is therefore a model comparison problem.

For any fixed value of $K \in \{0, 1, 2, \dots\}$, the model specification in Sections 4–7 is complete. The case $K = 0$ is a special case: it sets $\mathbf{\Sigma}_a = \mathbf{\Psi}_a$ (diagonal player covariance), corresponding to the assumption that all player-level cross-feature correlation is zero. This *diagonal model* serves as a natural baseline; the gain in predictive performance relative to it quantifies the value of modelling shared player-level covariance structure.

Whether shared player-level covariance is worth modelling at all is answered by the posterior-predictive check reported in Chapter 5 (§3), which compares model-implied and observed cross-feature correlations under $K = 0$ and $K \geq 1$ directly, without appeal to cross-validation; the choice of rank among $K \geq 1$ is argued qualitatively, as follows.

8.1. A qualitative argument for the rank. One caveat is worth flagging briefly before making that argument: the rank $K^* = 3$ selected here is not expected to match the eleven-factor count found in Chapter 3’s exploratory analysis. The two describe different things, for reasons given in full in Remark 4.19 below.

The rank actually adopted for the production model is selected on qualitative, structural grounds instead: the adequacy of each fit’s own residuals, and the extent to which the shared player-effect covariance Σ_a is still adding well-identified structure as K increases. This mirrors standard practice in exploratory and confirmatory factor analysis, where the number of retained factors is routinely set by fit diagnostics and interpretability rather than by a single automated rule, and is consistent with the professional decision making process in my day-to-day job: that a qualitative argument, grounded in residuals and in the variance structure of $\Lambda\Lambda^\top$, is an appropriate and sufficient basis for closing the choice of K .

REMARK 4.18 (Anchoring to the posterior mean, not to per-draw estimates). Both criteria below require the eigendecomposition of the common-variance matrix $\Lambda\Lambda^\top$ introduced in Section 10. Proposition 4.23 shows that, *for a fixed posterior draw*, this eigendecomposition is invariant to the identification constraint used to fit Λ . It is not, however, invariant *across* posterior draws in the sense needed to build a credible interval directly: sampling variability in $\Lambda^{(s)}$ can rotate the leading eigenvectors within the subspace they span from one draw to the next, so naively comparing $\mathbf{V}^{(s)}$ across draws risks comparing different rotations of the same subspace rather than a stable estimate of one direction. To avoid this, the analyses below fix a single reference orientation—the eigendecomposition of the *posterior-mean* common-variance matrix $\bar{\Lambda}\bar{\Lambda}^\top$, where $\bar{\Lambda} = \frac{1}{S_{mc}} \sum_s \Lambda^{(s)}$ —and project each draw’s loadings onto this fixed reference before summarising uncertainty. The same construction gives comparable per-player credible intervals: replacing $\mathbf{V}^{(s)}$ by the fixed $\bar{\mathbf{V}}$ in (25) yields $\mathbf{f}_i^{(s)} = \bar{\mathbf{V}}^\top \Lambda^{(s)} \boldsymbol{\eta}_i^{(s)}$, used in Chapter 5 to report posterior credible intervals on individual players’ canonical scores.

Criterion A: variance share of the PCs of $\bar{\Lambda}\bar{\Lambda}^\top$: For a candidate rank K , the eigenvalue d_K of the *last* (smallest) canonical factor, expressed as a share of $\sum_{k=1}^K d_k$, measures how much additional common-variance structure the K -th dimension contributes. A last-component share close to the trailing eigenvalues of the full uniqueness spectrum indicates that dimension is not resolving substantive shared structure.

Criterion B: residual adequacy under the fit’s own $t(\hat{\nu})$ noise model: For each candidate fit, the standardised residuals $z_{np} = (y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}) / (\hat{\sigma}_{e,n,p} \sqrt{\hat{\nu}/(\hat{\nu} - 2)})$ are compared against the 99th-percentile threshold τ_{99} implied by $t(\hat{\nu})$, where $\hat{\sigma}_{e,n,p}$ is the fit’s own residual scale at observation n (Definition 4.8 for a minutes-scaled fit; $\hat{\sigma}_{e,n,p} = \hat{\sigma}_{e,p}$, constant in n , for a constant-scale fit). If the empirical exceedance rate across all $N \times P$ cells is at or below the nominal 1%, the residual distribution is adequate under the fitted noise model; this is a floor check on model misspecification, not a criterion that discriminates between two ranks that both pass it.

8.2. Selected rank. $K = 2$ and $K = 3$ below are evaluated on the production, minutes-scaled, fixed- $\phi = 1/2$ family; $K = 4$ is carried over from the constant-scale family (Decision 2 of the revision that corrected this chapter: refitting $K = 4$ under minutes-scaling was not undertaken, since the existing constant-scale $K = 4$ fit already required substantially relaxed sampler settings and multi-day runtime for reasons unrelated to residual scaling). The three numbers move only marginally between families. Applying

Criteria A and B across $K = 2, 3, 4$ gives the following picture. The last canonical factor's common-variance share declines from 30.7% at $K = 2$ to 10.4% at $K = 3$ to 7.1% at $K = 4$ —a continued but decelerating decline, not a sharp drop to a noise floor at any point (Criterion A). Mean residual exceedance under τ_{99} is 0.77%, 0.79%, and 0.79% at $K = 2, 3, 4$ respectively—all comfortably below the 1% floor and essentially flat, so residual adequacy holds at every rank examined and does not discriminate among them (Criterion B).

Taken together, this comparison does not produce a clean reversal at $K = 4$ the way it does at $K = 3$ relative to $K = 2$: Criterion A mildly favours stopping at $K = 3$ (the fourth factor's contribution, while still non-trivial, is smaller than the third's), while Criterion B is uninformative at this rank range and offers no separate support either way. Read honestly, choosing $K = 3$ over $K = 4$ rests on Criterion A alone, and amounts to a preference for the simpler model: the fourth factor's contribution is smaller, but not small enough to ignore. This is a much weaker result than the case for $K \geq 1$ over $K = 0$, or even the milder case for $K = 3$ over $K = 2$; both of those were clear-cut, this one is not. The honest summary is that the data do not sharply distinguish $K = 3$ from $K = 4$. $K = 3$ is nonetheless retained as the production rank $K^* = 3$, used throughout the remainder of this chapter and in Chapter 5: it is the more parsimonious of two ranks that the evidence does not clearly separate, and closing the model at a defensible, adequately-supported rank was judged preferable to an open-ended search for further, likely similarly ambiguous, comparisons at $K \geq 5$.

REMARK 4.19 (Comparability with the exploratory factor count of Chapter 3). Chapter 3 reports an eleven-factor solution from classical, maximum-likelihood factor analysis of the raw player–season feature matrix. It is natural to ask why the Bayesian model's rank, $K^* = 3$, is so much smaller. The two quantities are not estimates of the same object. The exploratory factor analysis of Chapter 3 is fit directly to the raw player–season observations $\mathbf{y}_n$, treated as exchangeable rows with no player/season structure: its factor count therefore describes the rank of the *total, unpartitioned* covariance of $\mathbf{y}_n$, in which any two features that co-vary for any reason shared stable player skill, a league-wide tactical shift common to a subset of seasons, or coincidental season-to-season drift—contribute jointly to the recovered factor structure. The additive decomposition of Definition 4.1, by contrast, first partitions this total covariance into player (Σ_a), season (Σ_b), and residual (Σ_e) components, and the rank K studied in this section governs only Σ_a : the covariance of what is stable about a player once season-level drift and observation-level noise have already been subtracted out. Since Σ_b is diagonal by construction (§4, justified there on identifiability grounds given only $S = 12$ seasons), any genuine cross-feature covariance that is specifically season-driven cannot be absorbed by Σ_b and is instead either left in the residual or folded into Σ_a 's estimate; either way, K is not required to reproduce the rank of a covariance matrix that additionally contains season-level and idiosyncratic sources of dependence. A smaller K for the player-only covariance than the raw factor count is therefore the expected outcome of conditioning on more structure, not evidence against either analysis. This reasoning is offered as an explanation for the direction of the discrepancy; it has not been separately validated (for instance, by fitting a raw factor analysis restricted to a player only, season averaged version of the data to check whether its factor count moves closer to three), and should be read as such.

9. Variance decomposition and intraclass correlation

A key inferential output of the model is a decomposition of the total variance of each feature into contributions from the player, season, and residual components. Under the additive model (1) and the mutual independence of $\mathbf{a}_i$, $\mathbf{b}_j$, and ε_n , the marginal variance of feature p in observation n is

$$(19) \quad \text{Var}(y_{np}) = [\Sigma_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np}),$$

where $[\Sigma_a]_{pp}$ is given by (3), and, for the production model's minutes-weighted Student- t residual, $\text{Var}(\varepsilon_{np}) = \sigma_{e,n,p}^2 \cdot \nu / (\nu - 2)$ by Corollary 4.9 — a quantity that depends on n through the observation's playing time m_n , not on p alone.

Because $\text{Var}(\varepsilon_{np})$ varies across observations of the same feature, so does $\text{Var}(y_{np})$: there is no longer a single total variance, or a single variance decomposition, per feature, only one per feature at each exposure level. The definition below resolves this by fixing $m_n = m_{\text{ref}}$, the reference exposure of Definition 4.8, at which the scaling factor $(m_{\text{ref}}/m_n)^\phi$ equals one and $\sigma_{e,n,p}$ reduces to $\sigma_{e,p}$. This recovers a single number per feature, interpretable as the variance decomposition for a player–season observation at the reference playing time.

DEFINITION 4.20 (Intraclass correlation coefficients). The player, season, and residual intraclass correlation coefficients for feature p , evaluated at the reference exposure $m_n = m_{\text{ref}}$, are, respectively,

$$(20) \quad \text{ICC}_{a,p} = \frac{[\Sigma_a]_{pp}}{[\Sigma_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

$$(21) \quad \text{ICC}_{b,p} = \frac{\sigma_{b,p}^2}{[\Sigma_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

$$(22) \quad \text{ICC}_{e,p} = \frac{\text{Var}(\varepsilon_{np})}{[\Sigma_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

satisfying $\text{ICC}_{a,p} + \text{ICC}_{b,p} + \text{ICC}_{e,p} = 1$ for all p .

The quantity $\text{ICC}_{a,p}$ measures the proportion of total variance in feature p that is attributable to stable differences among players. It is directly interpretable as the reliability of the feature as a measure of stable player style: features with $\text{ICC}_{a,p}$ close to one are dominated by stable player characteristics, while features with small $\text{ICC}_{a,p}$ are dominated by season-level context or observation-level noise. This framing of variance decomposition in terms of intraclass correlation follows [22].

REMARK 4.21 (Bayesian ICC). Because all variance parameters are sampled from the posterior, the quantities defined in Definition 4.20 are themselves posterior quantities. For each posterior draw, a value of $\text{ICC}_{a,p}$ can be computed from the sampled values of $\mathbf{\Lambda}$, Ψ_a , $\sigma_{b,p}$, $\sigma_{e,p}$, and ν , yielding a full posterior distribution over each coefficient. Posterior medians and 95% credible intervals are reported in Chapter 5. The ICC framing is preferred over the raw variance decomposition because it is scale-free and comparable across features with different marginal variances.

10. Canonical factor directions via spectral decomposition

10.1. Why the identification constraint does not yield interpretable directions. The LT-PD constraint of Definition 4.12 resolves the rotational non-identifiability by fixing a particular coordinate system in the K -dimensional factor space, but it does so in a way that is mathematically convenient rather than interpretable. Specifically, the first column of $\mathbf{\Lambda}$ is constrained to load only on features 1 through P with feature 1 carrying the positive diagonal; the second column is constrained to load on features 2 through P with feature 2 carrying the positive diagonal; and so on. This triangular structure is a consequence of zeroing out the upper triangle of $\mathbf{\Lambda}$, not of any reasoning about football-domain structure. A different ordering of the P features would produce a different fitted $\mathbf{\Lambda}$, even though the resulting Σ_a would be unchanged. The columns of the fitted LT-PD loading matrix therefore cannot be reported or interpreted as football archetypes.

10.2. Spectral decomposition of the common variance matrix. A rotation-invariant representation of the factor structure is obtained by working directly with the common variance matrix $\mathbf{C} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$, which is uniquely determined by the model regardless of which identification constraint was used.

DEFINITION 4.22 (Spectral post-processing). For each posterior draw $s = 1, \dots, S_{\text{mc}}$, let $\mathbf{\Lambda}^{(s)}$ denote the sampled loading matrix. Compute the common variance matrix

$$(23) \quad \mathbf{C}^{(s)} = \mathbf{\Lambda}^{(s)} \left(\mathbf{\Lambda}^{(s)} \right)^\top \in \mathcal{S}_P^+,$$

and perform its spectral decomposition

$$(24) \quad \mathbf{C}^{(s)} = \mathbf{V}^{(s)} \mathbf{D}^{(s)} \left(\mathbf{V}^{(s)} \right)^\top,$$

where $\mathbf{D}^{(s)} = \text{diag}(d_1^{(s)} \geq \dots \geq d_K^{(s)} \geq 0)$ and $\mathbf{V}^{(s)} \in \mathbb{R}^{P \times K}$ has orthonormal columns. The k -th column $\mathbf{v}_k^{(s)}$ of $\mathbf{V}^{(s)}$ is the k -th principal eigenvector of $\mathbf{C}^{(s)}$; the corresponding eigenvalue $d_k^{(s)}$ is the amount of player-effect common variance explained by the k -th canonical factor direction. The rotated factor score for player i in draw s is

$$(25) \quad \mathbf{f}_i^{(s)} = \left(\mathbf{V}^{(s)} \right)^\top \boldsymbol{\eta}_i^{(s)} \in \mathbb{R}^K.$$

PROPOSITION 4.23 (Invariance of the spectral decomposition). *The eigenvalues $\{d_k^{(s)}\}_{k=1}^K$ and the common variance matrix $\mathbf{C}^{(s)}$ are invariant to any orthogonal transformation of the loading matrix. Replacing $\mathbf{\Lambda}^{(s)}$ by $\mathbf{\Lambda}^{(s)}\mathbf{Q}$ for any $\mathbf{Q} \in O(K)$ yields the same $\mathbf{C}^{(s)}$ and hence the same spectral decomposition (24).*

PROOF. By Proposition 4.11, $(\mathbf{\Lambda}^{(s)}\mathbf{Q})(\mathbf{\Lambda}^{(s)}\mathbf{Q})^\top = \mathbf{\Lambda}^{(s)}\mathbf{\Lambda}^{(s)\top} = \mathbf{C}^{(s)}$. $\square$

Proposition 4.23 is the key justification for reporting the eigenvectors $\mathbf{V}^{(s)}$ and eigenvalues $\{d_k^{(s)}\}$ as the canonical factor structure: their values depend only on the posterior common variance $\mathbf{C}^{(s)}$, not on which identification constraint was used to fit $\mathbf{\Lambda}$.

REMARK 4.24 (Sign convention). Eigenvectors are defined up to sign: if $\mathbf{v}$ is an eigenvector of $\mathbf{C}^{(s)}$, so is $-\mathbf{v}$. A sign convention is applied draw-by-draw before computing posterior summaries: for each canonical factor k , the sign of $\mathbf{v}_k^{(s)}$ is chosen so that the feature with the largest absolute loading on that eigenvector has a positive loading. This convention ensures that the posterior mean of $\mathbf{v}_k^{(s)}$ reflects a coherent direction rather than cancelling between draws of opposite sign.

CHAPTER 5

Results

This chapter reports the results of fitting the model of Chapter 4 to the production La Liga player–season dataset described in Chapter 2. Unless stated otherwise, every number, table, and figure in this chapter comes from the minutes-scaled, fixed- $\phi = 1/2$, Student- t production fit at $K^* = 3$ (Definition 4.1, Σ_a as in Definition 4.3, residual scale as in Definition 4.8), the model whose specification, identification, and rank selection were argued for in Chapter 4. §3 confirms $K^* = 3$ directly on this family, alongside $K = 2$; §2 reports the diagnostic comparison against an earlier constant-residual-scale fit and against a fit with ϕ estimated rather than fixed, which is why numbers from those two fits appear there and only there.

1. Model diagnostics and convergence

Table 1 summarises posterior convergence for the production fit, grouped by parameter family. $\hat{R}$ and effective sample size are computed in the usual way from four parallel chains. Three of the 144 entries of Λ_a are excluded from the Λ_a row: under the lower-triangular, positive-diagonal constraint of §6.2 these entries are fixed at exactly zero and never sampled, so $\hat{R}$ is undefined for them (zero posterior variance) rather than poorly mixed.

Convergence is excellent for the scalar and per-feature parameters: ν , ψ_a , σ_b , and σ_e all have $\hat{R} \leq 1.005$ and minimum bulk-ESS in the hundreds to thousands. Λ_a itself mixes markedly less cleanly (worst $\hat{R} = 1.032$, 36 of the 141 free entries above the conventional 1.01 threshold), an identification artefact anticipated by Remark 4.14: the raw loadings under the LT-PD constraint are not themselves the object of substantive interest, so this is not by itself informative about whether the model has recovered a genuine, stable shared-covariance structure.

TABLE 1. Posterior convergence summary for the production fit, by parameter family. ν is the Student- t degrees of freedom (Definition 4.6); Λ_a , Ψ_a are the factor loadings and uniqueness (Definition 4.3); σ_b, σ_e are the season and residual scales. The $\Lambda_a \Lambda_a^\top$ row summarises all $P(P+1)/2$ unique entries of the rotation-invariant common-variance matrix (Definition 4.22); the d_1, d_2, d_3 row summarises its three eigenvalues, the quantities Criterion A and Table 4 actually depend on.

Parameter	N	ESS _{bulk} : median (min)	Max $\hat{R}$	$N(\hat{R} > 1.01)$
$\Lambda_a \Lambda_a^\top$ (all entries)	1176	603 (154)	1.025	12
d_1, d_2, d_3 (eigenvalues)	3	446 (216)	1.004	0
Λ_a	141	343 (75)	1.032	36
ν	1	2895 (2895)	1.000	0
ψ_a	48	1127 (546)	1.005	0
σ_b	48	1615 (1128)	1.001	0
σ_e	48	3983 (2548)	1.002	0

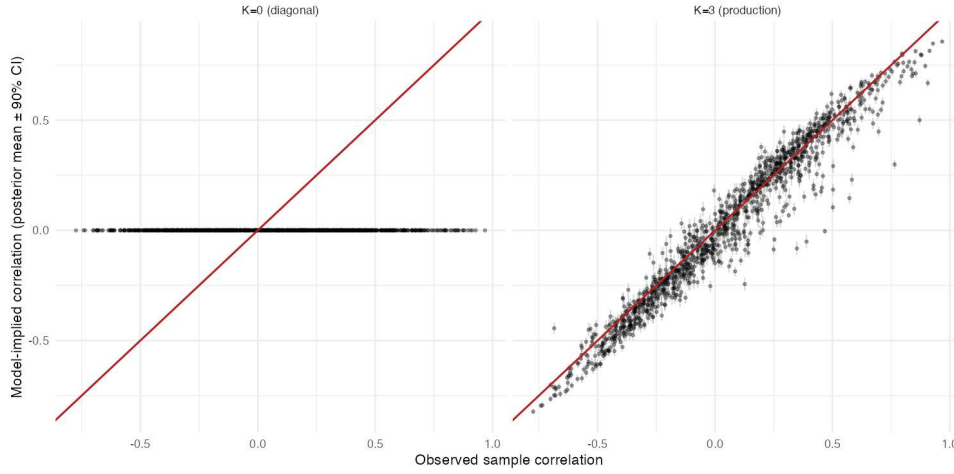


FIGURE 1. Posterior predictive check: observed pairwise feature correlations vs. the model-implied correlation $[\Sigma_a]_{pq} / \sqrt{\text{Var}(y_p)\text{Var}(y_q)}$ (Proposition 4.4), posterior mean $\pm 90\%$ credible interval across 4,000 draws. Left: $K = 0$, where Λ_a has no columns and every model-implied correlation is exactly zero by construction. Right: the production rank, $K^* = 3$. $r(\text{observed}, \text{model-implied}) = 0.975$ at $K^* = 3$.

Proposition 4.23 gives the rotation-invariant object that *is* of substantive interest: the common-variance matrix $\hat{\Lambda}_a \hat{\Lambda}_a^\top$. Rather than checking one representative entry, Table 1 reports convergence over all 1,176 of its unique entries (48 diagonal, 1,128 off-diagonal): median (minimum) bulk-ESS 603 (154), maximum $\hat{R} = 1.025$, with 12 of 1,176 entries (1.0%) above 1.01. That contrast is the point: the raw loadings look markedly worse than they did under the constant-scale fit, while the rotation-invariant object built from them looks, if anything, cleaner. Two things are worth saying about that last count, though. With 1,176 separate checks, a handful above the conventional 1.01 threshold is expected from multiple comparisons alone and is not, by itself, a sign of a mixing problem. $\hat{R}$ is also a less reliable diagnostic for entries whose posterior is concentrated near zero, which describes many off-diagonal entries at rank 3; restricting to entries with posterior-mean magnitude above 0.05 changes the picture only slightly, to 11 of 1,071 entries above 1.01, maximum $\hat{R} = 1.025$, so this caution does not materially change the read here. The second row of the table gives the same summary for the three eigenvalues d_1, d_2, d_3 of $\hat{\Lambda}_a \hat{\Lambda}_a^\top$ directly: bulk-ESS from 216 to 551, $\hat{R} \leq 1.004$ for all three, the cleanest read of convergence for the quantities Criterion A and the variance-share table below are actually built from.

Figure 1 checks whether the model reproduces the observed cross-feature correlation structure: since Σ_b and Σ_e are diagonal by construction, every model-implied off-diagonal correlation comes entirely from $\Sigma_a = \Lambda_a \Lambda_a^\top + \Psi_a$. The right panel's near-perfect agreement ($r = 0.975$) across all $\binom{48}{2} = 1,128$ feature pairs shows the shared factors, plus feature-specific uniqueness, are jointly sufficient to explain almost all of the pairwise dependence structure in the data. §3 returns to the left panel directly: it is the evidence that shared structure is not merely helpful but necessary.

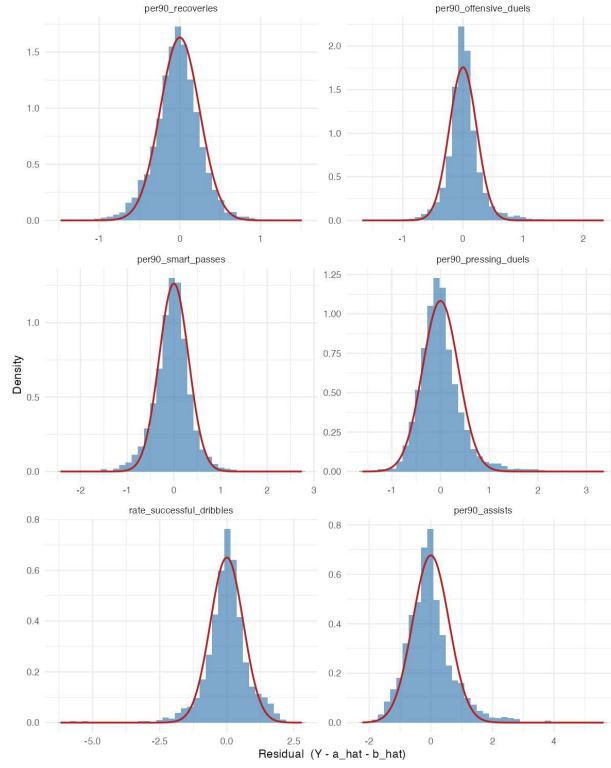


FIGURE 2. Residual check: $y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}$ against $\mathcal{N}(0, \hat{\sigma}_{e,p})$ for six features (top-2 by ICC_player, top-2 by ICC_season, bottom-2 by combined ICC; §4). The fitted noise model is Student- t , not Normal, so heavier-than-Normal tails here are expected, not a misspecification.

2. Resolving the observation-level variance scaling (ϕ)

Remark 4.10 states that the production model fixes $\phi = 1/2$. This section is where that choice is argued for, against a diagnostic fit with ϕ estimated and against the constant-scale specification both were built to correct. The motivation for scaling at all came from a direct diagnostic on the constant-scale fit: of the top-200 worst standardised-residual cells, 48.9% (88/180) came from the bottom quintile of playing time. An even split by minutes would predict 20%; the gap is not explainable by chance (binomial $p < 4.4 \times 10^{-18}$). Observations from low-minute player-seasons are markedly under-fit by a model that treats residual variance as constant regardless of exposure.

TABLE 2. Resolving ϕ : the constant-scale specification vs. two minutes-scaled variants. "Low-minute worst-cell share" repeats the diagnostic above, recomputed on each fit's own residuals.

Model	ν (90% CI)	ϕ (90% CI)	Low-minute share
Constant σ_e	4.90 [4.79, 5.01]	fixed at 0	48.9% (88/180)
$\phi = 0.5$ fixed (production)	5.67 [5.54, 5.80]	fixed at 0.5	8.3% (14/168)
ϕ estimated	5.98 [5.83, 6.13]	0.366 [0.360, 0.373]	8.3% (14/168)

Table 2 compares the constant-scale fit against two minutes-scaled refits: ϕ fixed at the Poisson-rate value of 0.5, and ϕ estimated freely. Both minutes-scaled variants eliminate the low-minute over-representation almost entirely, from 48.9% to 8.3%. Scaling addresses the diagnosed problem; it does not just reshuffle it. The estimated $\hat{\phi} = 0.366$ (90% CI [0.360, 0.373]) sits below 0.5. Residual noise therefore falls off more slowly with playing time than pure sampling error would predict, which is what you would expect if part of the extra variation in low-minute seasons is a genuine difference between players rather than measurement noise. Both scaled variants also push $\hat{\nu}$ higher, from 4.90 to 5.67–5.98: once exposure-driven heteroscedasticity is accounted for, less of the remaining spread needs to be attributed to heavy tails.

Verdict: minutes-scaling is necessary; the remaining question is whether ϕ should be fixed or estimated, and the production model fixes it at 0.5. It is the value predicted directly by Poisson-rate sampling theory, not a value read off this particular fit, so it does not risk fitting noise in the exponent itself. It also introduces one fewer free parameter than treating ϕ as unknown, at a cost the estimated fit’s own posterior suggests is small. Every subsequent result in this chapter comes from the minutes-scaled, fixed- $\phi = 0.5$ fit.

3. Rank selection: results

Two separate questions were posed in §8: whether low-rank Σ_a is worth having at all ($K = 0$ vs. $K \geq 1$), and, if so, which rank to use (K^* among $K \geq 1$). The evidence for the two questions is qualitatively different, and this section reports them separately rather than blending them. Both are now evaluated on the production family fixed in §2.

$K = 0$ vs. $K \geq 1$: decisive. The left panel of Figure 1 shows what a diagonal player covariance implies: every model-implied cross-feature correlation is exactly zero, by construction (Λ_a has no columns, and Σ_b , Σ_e contribute nothing off-diagonal either). Observed correlations span roughly -0.7 to $+0.9$ across the same 1,128 feature pairs. A model with $K = 0$ cannot reproduce this range under any parameter values; the right panel shows a $K \geq 1$ model can, closely. This is not a directional or approximate comparison: it follows from the structure of the $K = 0$ model itself, not from any diagnostic that could be read as inconclusive. Shared player-level covariance structure is real and worth modelling.

Choosing K^* among $K \geq 1$: qualitative, per §8.1. Table 3 recaps the two structural criteria across $K = 2, 3, 4$ that §8.1 defined. $K = 2$ and $K = 3$ are evaluated on the minutes-scaled, fixed- $\phi = 1/2$ family fixed above; $K = 4$ is not refit under that family (the existing constant-scale $K = 4$ fit is orders of magnitude more expensive to sample than $K = 2$ or $K = 3$, for reasons unrelated to minutes-scaling), so its row is carried over from the constant-scale family and reported as such rather than silently mixed in.

Criterion A mildly favours stopping at $K = 3$ (the fourth factor’s 7.1% share is smaller than the third’s 10.4%, but the decline from $K=2 \rightarrow 3 \rightarrow 4$ is continuous, not a sharp drop to a noise floor). Criterion

TABLE 3. Criteria A and B across candidate ranks (§8.1). Criterion A: last canonical factor’s share of $\bar{\Lambda}_a \bar{\Lambda}_a^\top$ ’s total variance. Criterion B: mean fraction of standardised residual cells exceeding τ_{99} , the 99th percentile of each fit’s own $t(\hat{\nu})$ (each fit’s estimated degrees of freedom $\hat{\nu}$ therefore differs, but is not shown as a separate column here). $K = 4$ ’s row is from the constant-residual-scale family (see text); $K = 2$ and $K = 3$ are from the production, minutes-scaled family.

K	Last-PC share of $\bar{\Lambda}_a \bar{\Lambda}_a^\top$	Mean $\Pr(z > \tau_{99})$
2	30.7%	0.77%
3	10.4%	0.79%
4	7.1%	0.79%

B is flat and uninformative at this range (all three ranks comfortably under the 1% adequacy floor). Read honestly, the case for $K = 3$ over $K = 4$ rests on Criterion A alone, and amounts to a preference for the simpler model under a still-shrinking-but-not-negligible fourth factor, not a clean reversal of the kind that separates $K = 0$ from $K \geq 1$. $K^* = 3$ is retained on parsimony grounds and used throughout the rest of this chapter.

4. Variance decomposition: intraclass correlations

Definition 4.20 defines $ICC_{a,p}$, $ICC_{b,p}$, $ICC_{e,p}$ as the share of feature p 's total variance attributable to player, season, and residual, respectively, evaluated at the reference exposure $m_n = m_{ref}$: under the production model's minutes-weighted residual scale the residual variance depends on playing time, so this is the variance decomposition for a player–season observation at the reference minutes figure, not an average over observations. Remark 4.21 notes these are themselves posterior quantities, since every variance component is sampled; Figure 3 plots the posterior-median split for the five features with the highest player ICC and, separately, the five features with the highest season ICC (the full 48-feature table is given in Table 1, Appendix A).

Player identity dominates the left-panel features: all five exceed $ICC_a = 0.85$ (recoveries, touch in box, forward passes, offensive duels, crosses), with essentially no season component. The right panel tells a complementary story: season ICC reaches 0.44 for smart passes and 0.30 for pressing duels, features that do not otherwise stand out on the player-dominance ranking. This is the quantitative counterpart to the qualitative claim motivating the whole model (§4): football performance profiles are substantially stable properties of the player for most features, but a handful of specifically tactical or rule-sensitive features carry a real, non-trivial season signal. §5 returns to smart passes' season trend directly.

This split is only available because the model gives player and season effects their own covariance structures from the outset, rather than fitting a single factor matrix to the pooled observations the way a classical, frequentist factor analysis does (Chapter 3). A one-shot analysis of the raw player–season rows has no player or season index to condition on, so any covariance between two features, whether it comes from stable player skill or a league-wide shift concentrated in a few seasons, is folded into the same estimated structure with no way to separate the two afterward. Here the low-rank factor structure of Definition 4.3

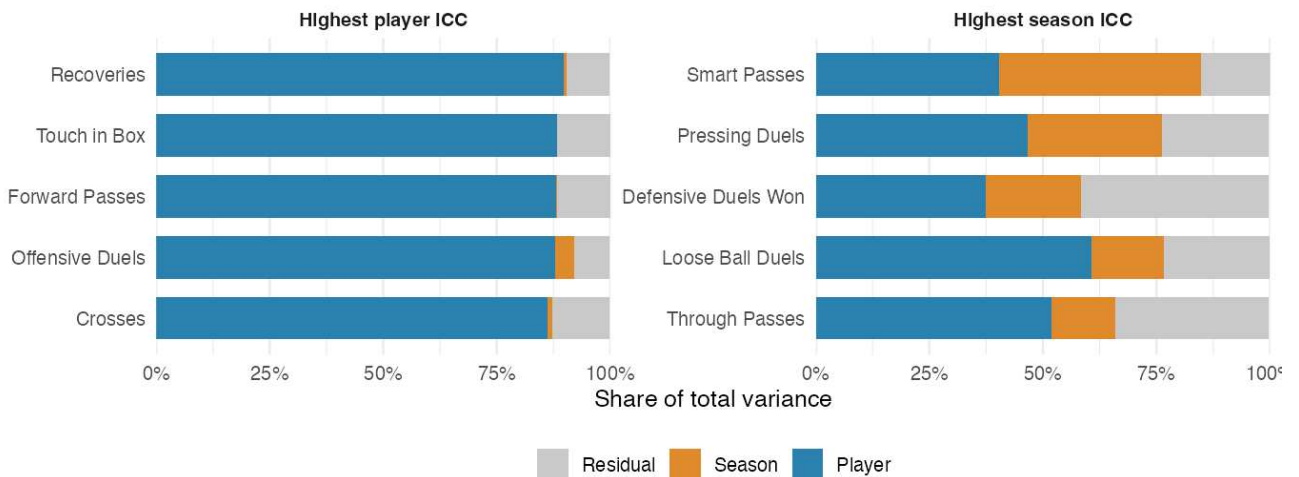


FIGURE 3. Player/season/residual variance share (posterior median) for the five features with the highest player ICC (left) and the five features with the highest season ICC (right).

applies specifically to Σ_a , so the ICC decomposition above falls directly out of that separation, not out of anything recoverable from a single fitted covariance matrix.

5. Season profiles

Consistent with Figure 3, season ICC is small for almost every feature, but small does not mean zero, and group-averaged season effects reveal coherent league-wide drift over the twelve seasons in the data (Figure 4).

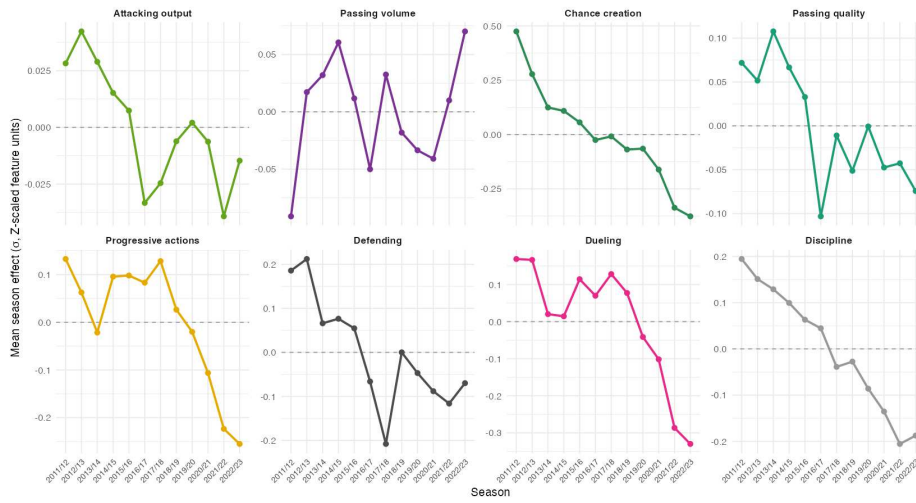


FIGURE 4. Group-averaged posterior-mean season effects $\hat{\mathbf{b}}_j$, in Z-scaled feature units, one panel per feature group, 2011/12–2022/23. Most groups drift downward from around 2017/18 onward.

Chance creation—the creative, line-breaking side of the passing game (smart passes, key passes, crosses, shot assists, through balls; see Section 6)—shows the cleanest and largest swing of any group in Figure 4: a high in exactly 2011/12–2012/13, the seasons spanning the end of Pep Guardiola’s Barcelona and Tito Vilanova’s title-winning continuation of it, followed by an almost uninterrupted decline for the rest of the sample. Dueling follows a related but less clean pattern: elevated in the same two seasons, it dips through the mid-2010s and recovers to a second, comparable high around 2017/18 before falling away sharply. Passing volume—bulk, directional ball circulation rather than creative passing—tells a different story: it is at its lowest, not its highest, in 2011/12, and only climbs back to its sample peak at the very end of the window (2022/23). Read together, the split argues that what was actually elevated in the Guardiola/Vilanova era was purposeful, creative passing specifically, not passing volume in general, which is exactly the distinction the group split in Section 6 was designed to expose. Passing quality and Progressive actions move less cleanly season to season but both sit within a broadly elevated regime through the mid-2010s, a period that includes Luis Enrique’s Barcelona (2014/15–2016/17): still dominant, but built more around the pace and individual quality of Messi, Suárez, and Neymar in transition than Guardiola’s extended possession sequences, a real, partial step away from pure tiki-taka before the league-wide decline properly sets in. Figure 4 is a league-wide average across every club, not Barcelona’s own numbers, so no single team’s history explains it on its own; but the timing matches the qualitative story football writers were already telling about this period, and the decline visible in most panels from around 2017/18 onward is consistent with a broader shift toward faster, more direct, transition-oriented football across La Liga later in the decade.



FIGURE 5. The six features with the largest peak-to-trough range in $\hat{\mathbf{b}}_j$ across seasons. Smart passes shows the largest range (2.4σ) and, uniquely among the six, a monotonic falling trend rather than a peak-and-recovery shape.

Smart passes stands out not just for its range but for its shape: every other feature in Figure 5 rises, falls, and partially recovers within the sample window, while smart passes falls steadily from 2011/12 to 2022/23 with no recovery at any point. A through-ball attempted at a league-average rate in 2011/12 would be attempted well below that rate a decade later, holding player identity fixed. Read as a tactical signal rather than a statistical curiosity, this is evidence of a leaguwide shift away from risk-laden, line-breaking passing and toward lower-risk possession retention over the sample window, consistent with the broader decline visible in the Chance creation and Passing quality panels of Figure 4 (smart passes and through passes are themselves both Chance-creation features).

6. Canonical factor loadings by feature group

Raw $\mathbf{\Lambda}_a$, fitted under the LT-PD constraint of §6.2, is an identification device and should not be interpreted directly (Remark 4.14). Following Definition 4.22, canonical factor directions are instead obtained by eigendecomposing the common-variance matrix $\bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top$ (Proposition 4.23 shows this is invariant to the identification constraint used at fit time).

Roughly a quarter of total player-level variance sits in Ψ_a rather than the three shared factors (Table 4), a meaningful idiosyncratic component that §8 returns to directly.

Three things stand out from Figure 6 and Table 5, one per axis. Concrete player examples are deferred to §7, where Table 6 names the players at each pole.

Defensive engagement's split of the passing game is now visible directly in the feature groups themselves: bulk, directional passing-volume features (forward, long, and lateral passes) load with its defensive pole, while

TABLE 4. Variance shares of the three canonical factors, for the common-variance matrix $\bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top$ and for total player variance $\bar{\mathbf{\Sigma}}_a = \bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top + \bar{\mathbf{\Psi}}_a$. The gap between the two columns for each PC is the share absorbed by feature-specific uniqueness $\bar{\mathbf{\Psi}}_a$ instead of the shared factors.

Canonical factor	Share of $\bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top$	Share of $\bar{\mathbf{\Sigma}}_a$
PC1	61.9%	46.3%
PC2	27.7%	21.0%
PC3	10.4%	8.2%

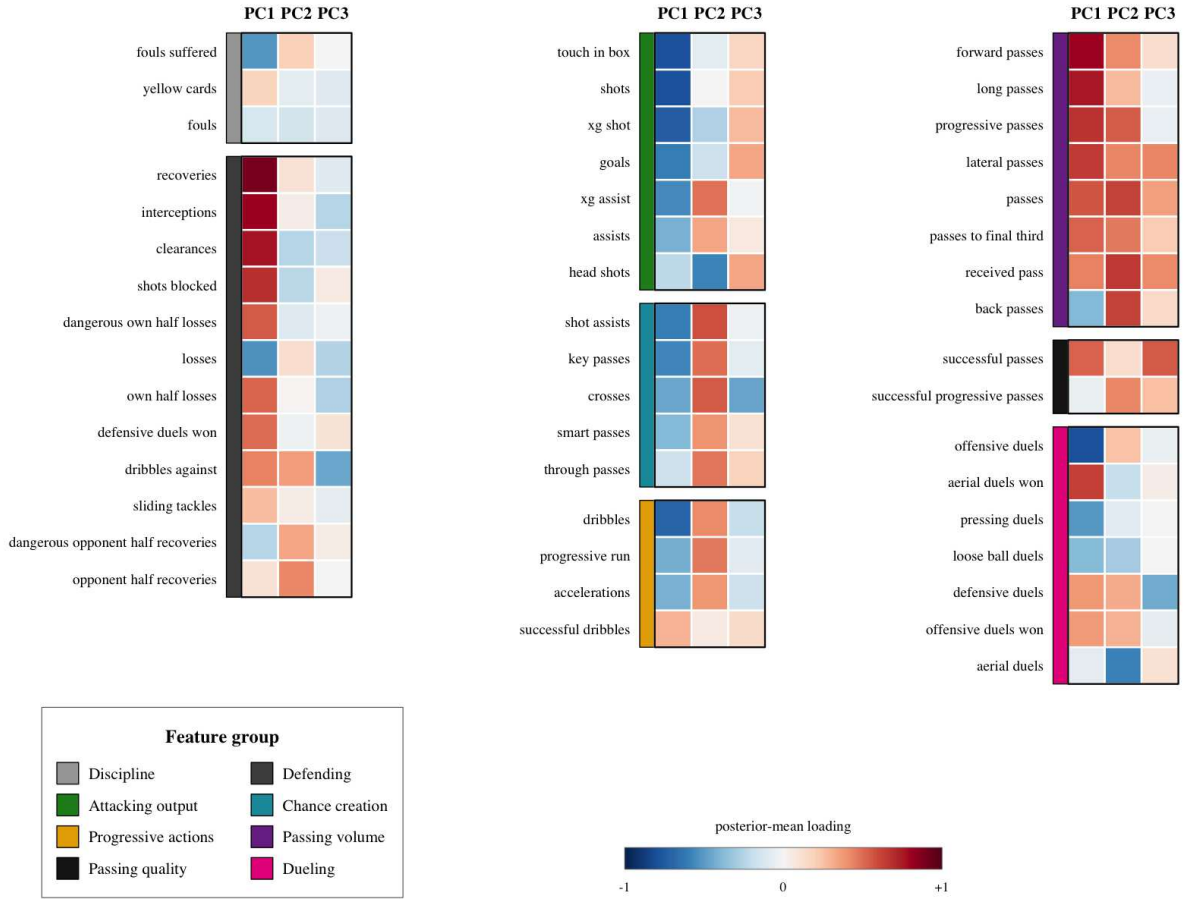


FIGURE 6. Posterior-mean loading of all 48 features on the three canonical factors – Defensive engagement, Technical orientation, and Possession retention – grouped into eight feature categories and coloured on a fixed $[-1, +1]$ diverging scale centred at zero. Each feature group's block of cells is bordered as a unit, with the group's colour shown as a strip to its left. All but seven of the 144 cell-loadings shown have a 90% credible interval excluding zero.

TABLE 5. The three canonical factors compared: a working definition and the four largest reliable loadings on each pole. Loadings are posterior means; all sixteen values shown have a 90% credible interval excluding zero (Figure 6).

	PC1	PC2	PC3
<i>Name</i>	Defensive engagement	Technical orientation	Possession retention
<i>Direction</i>	← 0 → attacking defensive	← 0 → aerial technical	← 0 → crossing retention
<i>Definition</i>	Recovers a player's defensive/build-up volume against their attacking output.	Separates ball-circulation play from aerial and physical duels.	Separates clean possession and direct threat from wide delivery and defensive workrate.
<i>Positive loadings</i>	recoveries (0.90), interceptions (0.81), forward passes (0.80), clearances (0.77)	received pass (0.65), back passes (0.62), passes (0.62), shot assists (0.58)	successful passes (0.55), lateral passes (0.43), received pass (0.42), passes (0.36)
<i>Negative loadings</i>	offensive duels (-0.80), touch in box (-0.80), shots (-0.79), expected goals per shot (-0.74)	aerial duels (-0.60), head shots (-0.58), loose ball duels (-0.29), expected goals per shot (-0.26)	crosses (-0.47), dribbles against (-0.46), defensive duels (-0.44), own-half losses (-0.26)

the Attacking passing group (key passes, smart passes, crosses, shot assists) loads with its attacking pole, a split by purpose rather than by volume. Technical orientation is largely independent of it: a target forward and a deep-lying passer can sit at a similar point on the defend/attack axis while landing at opposite ends of this one. Possession retention does not match a simple wide-versus-direct-play description, hence its “retention vs. crossing/defending” label: its positive pole combines possession quality with direct goal threat, and its negative pole mixes wide attacking play with contested defensive duels rather than sitting cleanly on either side alone. It adds information the other two do not carry: two attacking players who agree on both can still differ sharply in whether their output comes through retained possession and clean finishing or through crossing into contested situations.

7. The canonical style space

Projecting each player's posterior-mean effect onto the three canonical directions gives every player a position in a 3-dimensional style space. Figure 7 shows this space directly: k -means clustering ($k = 5$) on the Defensive engagement/Technical orientation plane recovers five groups that align closely with recognisable positional roles once ordered along the first of those two axes, from Forwards at the most attacking end to Centre backs at the most defensive.

Table 6 complements this population view with the most extreme individual players on each canonical factor, four per pole, chosen as the highest-magnitude players.

One name recurs more than any other in Table 6: Messi anchors an extreme pole on all three axes at once, Defensive engagement's attacking pole, Technical orientation's technical pole, and Possession retention's retention pole, more than any other player in the table. Because the three axes are constructed to be orthogonal, this is not a statistical inevitability: a player extreme on one axis carries no obligation to be extreme on the others. That Messi is exactly this rare, triple-extreme case is part of why he opens the case studies of §8.

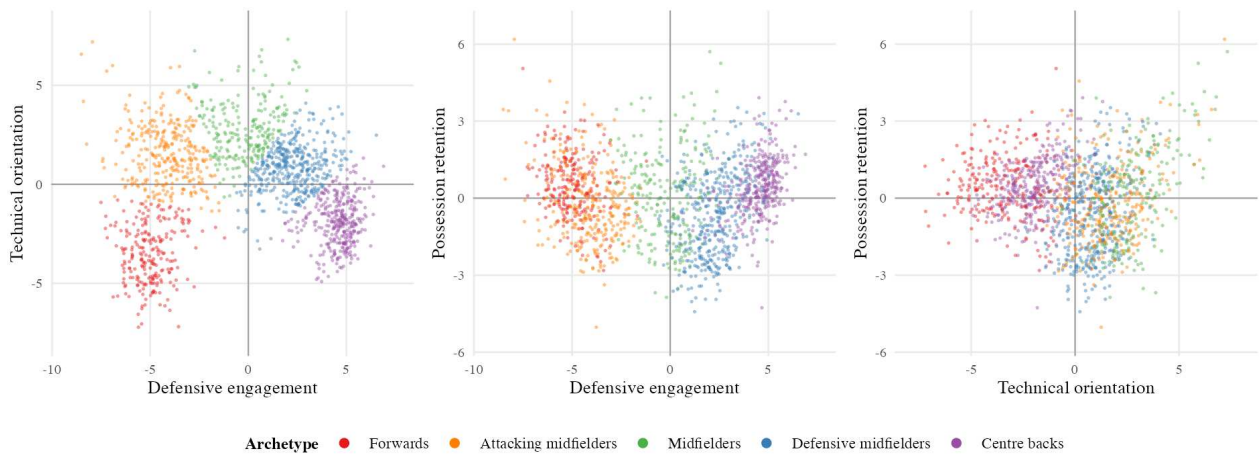


FIGURE 7. All 1,529 players' posterior-mean canonical scores, coloured by a five-cluster k -means partition of the Defensive engagement/Technical orientation plane (Forwards through Centre backs, ordered by the first of those two axes); the same colouring is carried into the other two projections without re-clustering. The clusters stay visually coherent wherever Defensive engagement is one of the two axes shown, since it alone still separates them, but collapse into a single overlapping cloud on the Technical orientation/Possession retention plane, the one projection that leaves it out.

TABLE 6. The four highest-magnitude, recognisable players on each pole of each canonical factor.

Defensive engagement		Technical orientation		Possession retention	
<i>Defensive</i>		<i>Technical</i>		<i>Retention</i>	
J. Mascherano	6.53	Xavi	7.32	L. Messi	6.19
Gerard Piqué	6.17	L. Messi	7.19	Xavi	5.71
Víctor Ruíz	6.00	Andrés Iniesta	6.79	M. Pjanić	5.25
S. Umtiti	5.91	J. Rodríguez	6.74	Cristiano Ronaldo	5.05
<i>Attacking</i>		<i>Aerial</i>		<i>Crossing</i>	
Neymar	-8.51	C. Stuani	-7.21	Iván Alejo	-5.03
Vinícius Júnior	-8.41	A. Budimir	-6.57	Juan Iglesias	-4.43
K. Mbappé	-8.24	Y. En-Nesyri	-6.15	O. Vranješ	-4.27
L. Messi	-7.95	S. Okazaki	-6.13	Houboulang Mendes	-4.03

8. Player similarity: measuring playing style

This section takes up, in depth, the question the Introduction posed at the outset: what does it mean for two players to be “similar,” and can that be measured directly, from what the model has already learned about a player’s style, rather than asserted from reputation alone?

8.1. Motivation: measuring style. The model’s own covariance structure already gives a principled way to define what “similar” means. Recall (Definition 4.3) that the player effect decomposes as

$$\mathbf{a}_i = \mathbf{\Lambda}_a \boldsymbol{\eta}_i + \boldsymbol{\psi}_a \odot \mathbf{z}_{u,i},$$

a *shared* component driven by the $K^* = 3$ canonical factors, and an *idiosyncratic* component that those factors do not explain. Style is read off the shared component: a player’s position in the 3-dimensional canonical factor space, $\mathbf{f}_i$ (Definition 4.22), the same coordinates used for the archetype clustering of §7, evaluated at its posterior mean throughout this section.

Cristiano Ronaldo makes the point concretely. He is a genuine outlier in exactly the way his reputation suggests: 5th most extreme player of 1,529 on Defensive engagement’s attacking pole and 4th most extreme on Possession retention’s direct-goal-threat pole, the same company as Messi on both axes. His single closest style-neighbour under this metric is Karim Benzema, a forward built the same way, direct and high-value in attack rather than a wide creator or a deep passer. What that match does *not* claim is that Benzema scored the way Ronaldo did over this sample: nothing about how much either player scored enters $\mathbf{f}_i$, by construction. Matching someone’s shared style position is exactly what a nearest style-neighbour means; matching every idiosyncratic thing that makes a generational player singular is a different, much harder claim this chapter does not make. Whether a genuinely well-defined production score belongs alongside style similarity, and how to build one that is not just a raw goals count, is future work (§10).

8.2. The similarity score. Style distance, $d(i, j) = \|\bar{\mathbf{f}}_i - \bar{\mathbf{f}}_j\|$ (Euclidean, on canonical scores standardised across all 1,529 outfield players), is the raw quantity behind every comparison in this section: smaller means more similar. On its own, though, a raw distance is hard to read at a glance, 0.65 or 6.8 mean little without a reference point, so this chapter also reports a normalised **similarity score**,

$$\text{Similarity}(i, j) = 100 \times \exp(-d(i, j)/\bar{d}) \%,$$

where $\bar{d}$ is the mean pairwise style distance across the whole population ($\bar{d} \approx 2.25$): exactly 100% only for an identical player ($d = 0$), decaying smoothly as distance grows, calibrated so that two players a population-typical distance apart score roughly 37%. The transform is monotone in $d(i, j)$, so it reorders nothing, it only rescales the same distance onto an interpretable 0–100% range. Table 7 and the forest-plot cards below report both numbers side by side.

8.3. Case studies: four targets. The remainder of this section works through four named players: Messi, Xavi, Sergio Ramos, and Cristian Stuani. Stuani specifically was chosen to probe a profile the other three do not touch: a recognised, prolific target forward built around aerial ability and physical presence, a combination none of Messi (technical dribbler), Xavi (deep-lying passer), or Ramos (defensive centre-back) represents. A secondary reason for all four is that their canonical scores span real extremes of the space, more completely than a one-player-per-axis account would suggest, as each subsection below details for its own target.

Between them, the four targets anchor five of the space’s six poles: Defensive engagement’s attacking side (Messi, Stuani) and defensive side (Ramos), Technical orientation’s technical side (Messi, Xavi) and aerial side (Stuani), and Possession retention’s positive side (Messi, Xavi, Ramos), with its negative side left uncovered rather than filled by reaching for a fifth name. That spread is why Table 7 and Figure 8 show four such different neighbourhoods below: each target sits in a different corner of the space, not variations on one profile.

TABLE 7. Each target's three closest style-neighbours, by style distance and similarity score (§8).

Target	Similar player	Style distance	Similarity score
L. Messi	Neymar	1.818	45%
	Pedro	2.140	39%
	Philippe Coutinho	2.172	38%
Sergio Ramos	Gerard Piqué	0.158	93%
	S. Umtiti	0.306	87%
	Javi García	0.308	87%
Xavi	M. Pjanić	0.647	75%
	T. Kroos	1.116	61%
	Andrés Iniesta	1.229	58%
C. Stuani	Manucho Gonçalves	0.561	78%
	E. Boateng	0.573	78%
	G. Carrillo	0.688	74%

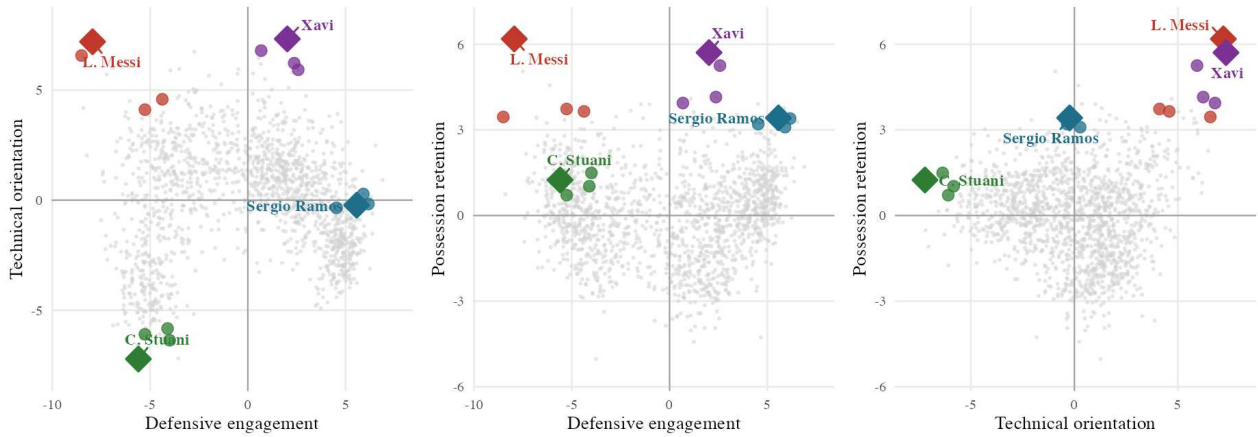


FIGURE 8. All 1,529 players' posterior-mean canonical scores (grey), with the four case-study players discussed in this section (diamonds) and their three closest style-neighbours (circles) in each of the three pairwise projections. Only the targets are labelled.

A three-dimensional style space is not honestly summarised by its first two coordinates, and Figure 8 shows why with the four players it highlights. On Defensive engagement alone, Ramos is a clear defensive extreme and Messi and Stuari are both clear attacking extremes on the opposite pole, while Xavi sits close to the population centre, a midfielder who is neither markedly defensive nor markedly attacking by that axis's measure. The Defensive engagement/Technical orientation panel breaks that grouping apart immediately: Messi and Xavi both score highly on the technical pole while Ramos sits near its centre, and Stuari anchors the opposite, aerial pole about as extremely as Messi and Xavi anchor the positive one — so the pair Defensive engagement groups together (Messi and Stuari, both attacking) splits completely on Technical orientation. The Technical orientation/Possession retention panel regroups them a third way: Messi and Xavi's points sit almost on top of each other, both far to the technical, high-retention end of both axes, while Ramos sits apart on Possession retention alone (a real, if less extreme, outlier there) and Stuari sits apart from all three, his extreme technical-orientation score dominating the panel regardless of his unremarkable retention value. No pair of axes groups these four players the same way twice; that is itself one of the more concrete illustrations in this chapter of why the third axis earns its place rather than being a minor correction to the first two.

The three most distant players per target (grey squares in the forest-plot cards below) are chosen from the top decile of style distance from that target, as the three most recognisable names in that decile, capped at not reaching past roughly its top 30 players by raw distance; the rule is applied identically to all four targets. Each subsection below reads off what that rule actually returns for its own target, and whether it makes sense.

8.3.1. *Messi: no one like him.* Messi is the one player in this chapter extreme on all three axes at once (§7): 4th most extreme on Defensive engagement's attacking pole, 3rd on Technical orientation's technical pole, 1st, the single most extreme value in the dataset, on Possession retention's retention pole. On the pitch that reads as a player who combines a very high volume of attacking involvement with technical, low-risk possession play and the most efficient direct scoring threat in the league at once, rather than trading one of those off for another the way most attacking players do.

Table 7 and Figure 8 show what that combination costs him: Neymar, Pedro, and Philippe Coutinho are his three closest style-neighbours, but even Neymar, the closest, only scores 45%, the lowest top-match similarity of any of the four targets. His three most distant players, O. Vranješ, Juan Iglesias, and Iván Alejo, are low-minute fringe players rather than rival attacking talents, single-digit similarity scores across the board: his profile is extreme enough that its opposite is occupied by nobody in particular, not by anyone whose game specifically inverts his. Figure 9's Defensive engagement and Possession retention intervals for the three neighbours overlap Messi's own comfortably even so, while the three distant players sit clearly apart on every axis.

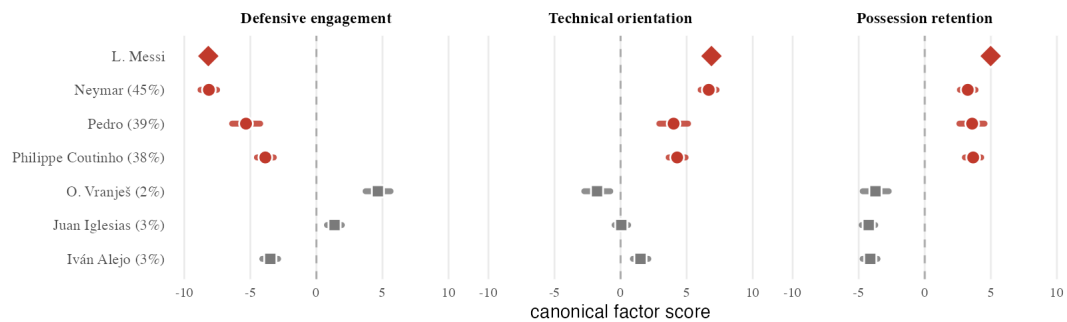


FIGURE 9. Messi: posterior 90% credible intervals on all three canonical factors for Messi, his three style-neighbours, and his three most distant players (grey), each labelled with its similarity score. Own scale; not comparable in absolute terms to the other three targets' cards.

8.3.2. *Xavi: the pinnacle of possession football.* Xavi is Technical orientation's defining case: the single most extreme value in the dataset on its technical, ball-circulation pole, and a real Possession retention extreme too (2nd most extreme, the retention pole), while sitting close to the population centre on Defensive engagement. That is a deep-lying playmaker read directly off the axes: not a volume-defined attacker or defender at all, a player whose entire footballing identity is retaining and progressing the ball better than almost anyone else in the league.

His three closest style-neighbours, Miralem Pjanić, Toni Kroos, and Andrés Iniesta (Table 7), are three players built around exactly that: retaining and circulating possession, matching that axis's description directly, with similarity scores from 75% down to 58%, the strongest top-three spread of any target here. His three most distant players, Budimir, Larin, and En-Nesyri, are all aerial-target forwards, its opposite pole, at single-digit similarity. Figure 10's technical-orientation panel makes the contrast visually explicit: Xavi's cluster and the three forwards' do not overlap at all.

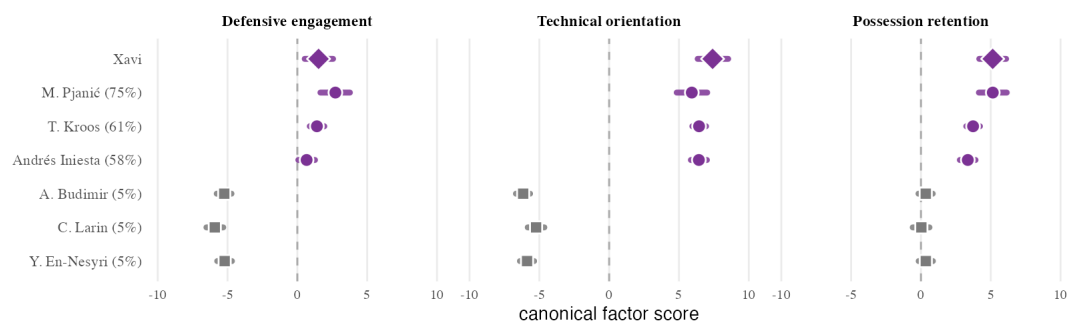


FIGURE 10. Xavi: posterior 90% credible intervals on all three canonical factors for Xavi, his three style-neighbours, and his three most distant players (grey), each labelled with its similarity score. Own scale.

8.3.3. *Ramos: the art of defending.* Ramos anchors Defensive engagement's defensive pole (138th most extreme, a real outlier though a quieter one than Messi's attacking-side extreme), sits close to the population centre on Technical orientation, and is also, somewhat unexpectedly, a Possession retention extreme (39th). On the pitch: a genuine defensive-volume outlier who is not a one-dimensional specialist, his position on the technical axis rules out being purely aerial/physical or purely technical, and his position on the retention axis adds a real, if quieter, quality not usually associated with a centre-back's profile.

Gerard Piqué is Ramos's clear closest style-neighbour (93%), ahead of a near-tie between S. Umtiti (87%) and Javi García (87%) for second and third, consistent with the well-known Ramos–Piqué pairing in football conversation. One of Ramos's three most distant players is Messi himself, the same defensive-engagement contrast this chapter returns to repeatedly, restated here as a distance rather than a pair of separate extremes. Figure 11 shows what the point estimates alone do not: on that axis, Ramos's own interval overlaps all three neighbours, and Umtiti's and García's overlap each other closely, while Piqué's sits more clearly apart, the visual confirmation of his standing as the sharper match.

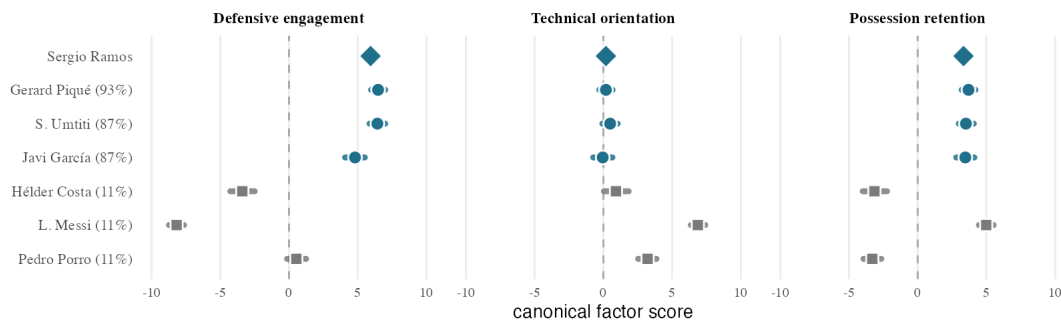


FIGURE 11. Ramos: posterior 90% credible intervals on all three canonical factors for Ramos, his three style-neighbours, and his three most distant players (grey), each labelled with its similarity score. Own scale.

8.3.4. *Stuani: the definition of a 9.* Stuani is a clear Defensive engagement extreme (129th, the same attacking pole as Messi) and, more strikingly, the 2nd most extreme value on Technical orientation in the entire dataset, anchoring its aerial/physical pole about as completely as Xavi and Messi anchor its opposite; he is unremarkable on Possession retention, the one axis his profile does not touch. That is a classic target forward read directly off the axes: physical, aerial, direct in attack, with none of the ball-retention or creative-passing quality that defines that axis's positive pole.

His three closest style-neighbours, Manucho Gonçalves, E. Boateng, and G. Carrillo (similarity scores 78%–74%), are journeyman target forwards, not other recognised strikers of comparable stature, a real finding about his position in the space rather than a data-quality artefact: style distance alone has no way to separate a prolific aerial striker from a similarly-built one who produces far less, and that is exactly the gap a future performance score (§10) would let this chapter close instead of gesture at. His three most distant players are, strikingly, Xavi, Messi, and Andrés Iniesta, two of the other three case-study targets and one of the most celebrated passers of the same generation, all drawn from the same short list of players who anchor the technical-orientation axis's technical pole: among the cleanest distant-player results the dataset produces, since the axis that motivated choosing Stuani is exactly the axis on which its opposite is most unambiguous. Figure 12 shows his own interval overlapping all three neighbours closely on both the defensive-engagement and technical-orientation axes, while the three distant players sit clearly on the opposite side of both axes with no overlap at all.

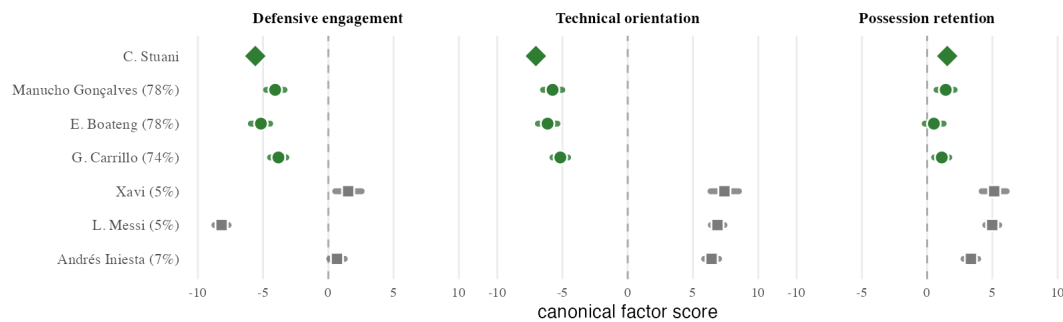


FIGURE 12. Stuari: posterior 90% credible intervals on all three canonical factors for Stuari, his three style-neighbours, and his three most distant players (grey), each labelled with its similarity score. Own scale.

9. Applications: scouting and recommendations

Clubs field player requests through agents and recruitment platforms constantly, filtered on criteria the model here says nothing about: position, age, contract status, market value. None of those filters says anything about *how* a player plays. Wanting “a Xavi” on a budget that will never buy the genuine article is not an edge case, it is one of the most common requests a recruitment department makes, and answering it well means exactly the separation this chapter builds: style, measured directly from playing data, independent of reputation or price. Recruitment in practice is not a matter of sorting by a single overall rating and taking the top name at a position; the useful question is almost always which of the affordable options plays in the right way.

The recipe follows directly from §8: fix a target profile, take a candidate pool already filtered on the hard, non-style criteria a club actually needs, rank what remains by similarity score, and return a scored, ranked shortlist. A club will not get a player who performs at the target’s level from this alone, that half of the question is a genuine performance score, taken up as future work below, but it is considerably more likely to find a player who *plays like* the target than by working from reputation or price tier alone. Being similar in style does not require being as good.

This is a style-matching tool, not a recruitment decision system: market value, injury history, contract situation, and league context beyond what the model already conditions on are not modelled here and belong alongside its output as separate filters, not replacements for it. Within that limit it offers something a scout’s individual judgement, applied one remembered name at a time, cannot: a precise, low-dimensional style representation computed across an entire league at once. This kind of comparison already happens, by hand, every transfer window; a tool that does it directly from data, at this level of precision and across a whole population rather than a shortlist of remembered names, is not something readily available off the shelf.

10. Future work

Four directions were considered during this work and are reported here as scoped future work rather than as contributions of the present thesis.

A performance score alongside similarity. §8 deliberately measures style only: two players can share a high similarity score while producing very differently, Stuari’s case study being the clearest example. A genuinely useful companion is a performance score, defined carefully enough to survive scrutiny rather than a raw goals count (which conflates opportunity, minutes, and finishing quality) or any single loose per90

rate. Paired with the similarity score, it would let a shortlist be built in two steps instead of one: filter to players who play like the target, then rank that shortlist by this measure to surface the best of them, rather than asking one number to do both jobs.

Uncertainty beyond posterior means. The model is fully Bayesian throughout, but two summaries still reduce a full posterior to a single number: the archetype clustering of §7 assigns one hard k -means cluster per player rather than a soft, probability-weighted membership, and the similarity scores of §8 report one distance per pair rather than a credible interval on it. The first is a direct extension of the uncertainty already reported for canonical scores; the second is harder, since a pairwise distance’s uncertainty depends on the joint draw-by-draw behaviour of both players together, not either one’s marginal interval, and was not attempted here.

Distance-based latent-space models. Chapter 4’s factor-analytic model represents style through a shared covariance structure; a complementary alternative represents it directly through pairwise dissimilarities via Bayesian multidimensional scaling [18], with hierarchical extensions for repeated player–season observations [28]. Chapter 3’s classical MDS is a first, non-probabilistic step in that direction; a fully Bayesian version would attach posterior uncertainty directly to each player’s position rather than obtaining it indirectly through canonical factor scores.

Team-season effects and tactical fit. No team identifier exists anywhere in the current data pipeline, so a genuine player-vs-system comparison is not buildable without new data engineering. A team-season effect $c_{t(i,j),j}$ would separate manager and tactical-system effects from stable player identity and would be the natural foundation for a real tactical-fit score; the closest available proxy today is simply a player’s archetype cluster (§7).

CHAPTER 6

Conclusion

Football clubs and analysts routinely ask which players resemble each other in style, yet answer that question from memory and reputation rather than from a stated, checkable definition of similarity. This thesis set out to replace that informal judgement with a hierarchical Bayesian model of player-season performance capable of recovering a small number of interpretable stylistic dimensions and of quantifying how similar any two players are, together with the uncertainty in that judgement.

The evidence assembled across the preceding chapters supports that aim on its own terms. Exploratory analysis established that La Liga's 48 standardised performance features are neither redundant nor irreducibly high-dimensional: principal component analysis, factor analysis, and multidimensional scaling all pointed to genuine, moderate-dimensional structure, without settling on a single preferred representation. The additive hierarchical model developed afterward resolved that ambiguity by treating player style as a shared, low-rank component of player identity, separate from season-level drift and from residual variation whose heavy tails and minutes-dependence needed explicit modelling rather than being averaged away. The results chapter showed that this structure is not a modelling convenience: the fitted covariance reproduces the observed cross-feature correlations almost exactly, and three canonical axes, named directly from their own loadings as Defensive engagement, Technical orientation, and Possession retention, organise the entire outfield population into recognisable positional archetypes without positional labels ever entering the model.

Two results connect the statistical machinery back to the practical question the thesis opened with. First, the same covariance structure that identifies the three axes also defines a single, bounded similarity score, so that how similar two players are has one consistent answer regardless of which pair is asked about, rather than a different ad hoc comparison each time. Second, the four case studies show that this score behaves the way a useful measure should: it separates genuinely rare profiles (Messi, whose closest match is still only 45% similar) from tightly clustered ones (Ramos and Piqué, at 93%), and, in Stuardi's case, it returns neighbours who share his style but not, by reputation, his end product.

That last result is informative in its own right: a style-only measure should not be expected to recover output, and showing exactly where it stops is more useful than a measure that quietly conflates the two. It is also the most direct link to the further work identified alongside the results themselves, where a companion performance score, uncertainty on the similarity scores, and team-level effects are the natural next steps, each building on the representation established here rather than replacing it.

Beyond its use as a scouting tool, the contribution of this thesis is methodological: it demonstrates that a fully Bayesian, identifiable factor model can be fit to a real, moderately sized sports dataset, diagnosed rigorously, and translated into quantities, canonical axes, a similarity score, positional archetypes, that a football analyst, not only a statistician, can read directly. That combination of statistical accountability and domain interpretability is what a purely descriptive or a purely predictive approach to player data cannot offer on its own.

APPENDIX A

Full intraclass-correlation table

The full 48-feature player, season, and residual ICC table referenced from §4 is given below (posterior median with 90% credible interval), ordered by descending player ICC; Figure 3 highlights the ten most informative rows directly in the main text.

Table 1. Full player, season, and residual ICC (posterior median [90% CI]) for all 48 features, ordered by descending player ICC.

Feature	ICC(player)	ICC(season)	ICC(resid.)
recoveries	0.90 [0.89, 0.91]	0.01 [0.00, 0.02]	0.09 [0.09, 0.10]
touch in box	0.88 [0.87, 0.89]	0.00 [0.00, 0.00]	0.12 [0.11, 0.13]
forward passes	0.88 [0.87, 0.89]	0.00 [0.00, 0.01]	0.12 [0.11, 0.12]
offensive duels	0.88 [0.83, 0.90]	0.04 [0.02, 0.10]	0.08 [0.07, 0.09]
crosses	0.86 [0.85, 0.87]	0.01 [0.00, 0.03]	0.13 [0.12, 0.14]
passes	0.86 [0.84, 0.87]	0.01 [0.00, 0.02]	0.13 [0.12, 0.14]
long passes	0.85 [0.84, 0.86]	0.00 [0.00, 0.01]	0.15 [0.14, 0.16]
xg shot	0.85 [0.84, 0.86]	0.00 [0.00, 0.00]	0.15 [0.14, 0.16]
clearances	0.84 [0.82, 0.85]	0.02 [0.01, 0.04]	0.14 [0.13, 0.15]
shots	0.84 [0.83, 0.85]	0.01 [0.00, 0.01]	0.15 [0.14, 0.17]
lateral passes	0.83 [0.81, 0.84]	0.01 [0.01, 0.03]	0.16 [0.15, 0.17]
progressive passes	0.82 [0.78, 0.84]	0.04 [0.02, 0.09]	0.14 [0.13, 0.15]
aerial duels	0.80 [0.74, 0.83]	0.06 [0.03, 0.13]	0.14 [0.12, 0.15]
dribbles	0.80 [0.70, 0.85]	0.11 [0.06, 0.21]	0.09 [0.08, 0.10]
received pass	0.79 [0.75, 0.81]	0.04 [0.02, 0.09]	0.17 [0.15, 0.18]
interceptions	0.78 [0.72, 0.81]	0.06 [0.03, 0.14]	0.15 [0.14, 0.17]
back passes	0.78 [0.76, 0.79]	0.01 [0.00, 0.03]	0.21 [0.20, 0.23]
passes to final third	0.78 [0.76, 0.79]	0.02 [0.01, 0.04]	0.21 [0.19, 0.22]
successful passes	0.77 [0.74, 0.79]	0.03 [0.02, 0.07]	0.20 [0.18, 0.21]
shot assists	0.77 [0.75, 0.78]	0.01 [0.00, 0.02]	0.22 [0.21, 0.24]
fouls suffered	0.74 [0.71, 0.76]	0.02 [0.01, 0.05]	0.24 [0.22, 0.26]
progressive run	0.73 [0.69, 0.76]	0.04 [0.02, 0.09]	0.22 [0.20, 0.24]
opponent half recoveries	0.71 [0.69, 0.72]	0.01 [0.00, 0.01]	0.29 [0.27, 0.31]
head shots	0.69 [0.67, 0.71]	0.00 [0.00, 0.00]	0.31 [0.29, 0.33]
defensive duels	0.69 [0.61, 0.73]	0.10 [0.05, 0.20]	0.21 [0.19, 0.23]
goals	0.68 [0.66, 0.70]	0.00 [0.00, 0.00]	0.32 [0.30, 0.34]
aerial duels won	0.68 [0.66, 0.70]	0.01 [0.00, 0.02]	0.31 [0.30, 0.33]
key passes	0.66 [0.64, 0.68]	0.00 [0.00, 0.01]	0.33 [0.31, 0.35]
xg assist	0.66 [0.64, 0.68]	0.00 [0.00, 0.01]	0.33 [0.31, 0.35]
dribbles against	0.65 [0.56, 0.70]	0.14 [0.08, 0.27]	0.20 [0.17, 0.22]
losses	0.65 [0.58, 0.69]	0.10 [0.05, 0.19]	0.25 [0.22, 0.27]
fouls	0.62 [0.59, 0.65]	0.03 [0.02, 0.07]	0.34 [0.32, 0.37]
accelerations	0.61 [0.52, 0.66]	0.14 [0.08, 0.27]	0.25 [0.21, 0.27]
loose ball duels	0.61 [0.51, 0.66]	0.16 [0.09, 0.29]	0.23 [0.19, 0.26]
shots blocked	0.61 [0.58, 0.63]	0.01 [0.00, 0.02]	0.38 [0.36, 0.40]
sliding tackles	0.57 [0.50, 0.62]	0.12 [0.07, 0.23]	0.30 [0.26, 0.33]
through passes	0.52 [0.44, 0.56]	0.14 [0.08, 0.28]	0.34 [0.28, 0.37]
own half losses	0.51 [0.48, 0.54]	0.03 [0.01, 0.07]	0.46 [0.43, 0.48]
offensive duels won	0.49 [0.46, 0.52]	0.02 [0.01, 0.05]	0.49 [0.46, 0.52]

continued on next page

(Table 1 continued from previous page)

Feature	ICC(player)	ICC(season)	ICC(resid.)
pressing duels	0.47 [0.35, 0.54]	0.29 [0.19, 0.48]	0.24 [0.18, 0.28]
successful progressive passes	0.42 [0.37, 0.45]	0.10 [0.05, 0.20]	0.49 [0.43, 0.52]
dangerous own half losses	0.42 [0.39, 0.44]	0.01 [0.01, 0.04]	0.57 [0.54, 0.59]
smart passes	0.40 [0.28, 0.50]	0.44 [0.31, 0.61]	0.15 [0.11, 0.19]
defensive duels won	0.37 [0.30, 0.42]	0.21 [0.12, 0.36]	0.41 [0.33, 0.47]
dangerous opponent half recoveries	0.37 [0.34, 0.40]	0.01 [0.00, 0.02]	0.62 [0.59, 0.65]
yellow cards	0.37 [0.34, 0.39]	0.02 [0.01, 0.05]	0.61 [0.58, 0.64]
assists	0.36 [0.34, 0.39]	0.01 [0.00, 0.02]	0.63 [0.60, 0.65]
successful dribbles	0.30 [0.28, 0.33]	0.01 [0.01, 0.03]	0.68 [0.65, 0.71]

APPENDIX B

Full canonical factor loadings

Table 5 in §6 shows only the four largest reliable loadings per pole per canonical factor, for readability. The full list, every loading with a 90% credible interval excluding zero, is given below, grouped by feature category and sign.

Table 1. Every reliable (90% credible interval excluding zero) posterior-mean loading on each canonical factor, by sign and feature group. Table 5 in the main text shows only the four largest per pole; this is the full list underlying Figure 6.

Group	Loadings
	PC1: Defensive engagement
	<i>Positive loadings (defensive pole)</i>
Discipline	
Defending	• yellow cards (0.19)
Progressive actions	• recoveries (0.90) • interceptions (0.81) • clearances (0.77) • shots blocked (0.69) • dangerous own half losses (0.55) • own half losses (0.52) • defensive duels won (0.51) • dribbles against (0.44) • sliding tackles (0.26) • opponent half recoveries (0.11)
Passing volume	• successful dribbles (0.30)
Passing quality	• forward passes (0.80) • long passes (0.77) • progressive passes (0.68) • lateral passes (0.64) • passes (0.57) • passes to final third (0.53) • received pass (0.45)
Dueling	• successful passes (0.53) • aerial duels won (0.63) • defensive duels (0.38) • offensive duels won (0.37)

continued on next page

(Table 1 continued from previous page)

Group	Loadings
	<i>Negative loadings (attacking pole)</i>
Discipline	• fouls suffered (-0.52) • fouls (-0.13)
Defending	• losses (-0.54) • dangerous opponent half recoveries (-0.25)
Attacking output	• touch in box (-0.80) • shots (-0.79) • xg shot (-0.74) • goals (-0.61) • xg assist (-0.56) • assists (-0.42) • head shots (-0.22)
Chance creation	• shot assists (-0.61) • key passes (-0.58) • crosses (-0.46) • smart passes (-0.39) • through passes (-0.17)
Progressive actions	• dribbles (-0.72) • progressive run (-0.42) • accelerations (-0.42)
Passing volume	• back passes (-0.39)
Dueling	• offensive duels (-0.80) • pressing duels (-0.51) • loose ball duels (-0.38) • aerial duels (-0.07)
	PC2: Technical orientation
	<i>Positive loadings (technical pole)</i>
Discipline	• fouls suffered (0.20)
Defending	• opponent half recoveries (0.43) • dribbles against (0.36) • dangerous opponent half recoveries (0.33) • losses (0.13) • recoveries (0.10) • sliding tackles (0.05) • interceptions (0.05) • own half losses (0.01)
Attacking output	• xg assist (0.49) • assists (0.33)

continued on next page

(Table 1 continued from previous page)

Group	Loadings
Chance creation	• shot assists (0.58) • crosses (0.55) • key passes (0.50) • through passes (0.47) • smart passes (0.39)
Progressive actions	• progressive run (0.46) • dribbles (0.42) • accelerations (0.39) • successful dribbles (0.06)
Passing volume	• received pass (0.65) • back passes (0.62) • passes (0.62) • progressive passes (0.54) • passes to final third (0.46) • lateral passes (0.43) • forward passes (0.42) • long passes (0.27)
Passing quality	• successful progressive passes (0.42) • successful passes (0.14)
Dueling	• defensive duels (0.32) • offensive duels won (0.30) • offensive duels (0.25)
Discipline	<i>Negative loadings (aerial pole)</i> • fouls (-0.15) • yellow cards (-0.07)
Defending	• clearances (-0.25) • shots blocked (-0.23) • dangerous own half losses (-0.09) • defensive duels won (-0.03)
Attacking output	• head shots (-0.58) • xg shot (-0.26) • goals (-0.17) • touch in box (-0.07)
Dueling	• aerial duels (-0.60) • loose ball duels (-0.29) • aerial duels won (-0.19) • pressing duels (-0.08)
PC3: Possession retention <i>Positive loadings (retention pole)</i>	

continued on next page

(Table 1 continued from previous page)

Group	Loadings
Defending	• defensive duels won (0.10) • shots blocked (0.06) • dangerous opponent half recoveries (0.05)
Attacking output	• goals (0.34) • head shots (0.33) • xg shot (0.27) • shots (0.21) • touch in box (0.17) • assists (0.06)
Chance creation	• through passes (0.19) • smart passes (0.11)
Progressive actions	• successful dribbles (0.14)
Passing volume	• lateral passes (0.43) • received pass (0.42) • passes (0.36) • passes to final third (0.22) • back passes (0.15) • forward passes (0.13)
Passing quality	• successful passes (0.55) • successful progressive passes (0.25)
Dueling	• aerial duels (0.12) • aerial duels won (0.05)
	<i>Negative loadings (crossing pole)</i>
Discipline	• fouls (-0.11) • yellow cards (-0.09)
Defending	• dribbles against (-0.46) • own half losses (-0.26) • losses (-0.25) • interceptions (-0.25) • clearances (-0.18) • recoveries (-0.09) • sliding tackles (-0.06) • dangerous own half losses (-0.03)
Chance creation	• crosses (-0.47) • key passes (-0.07) • shot assists (-0.03)
Progressive actions	• dribbles (-0.19) • accelerations (-0.17) • progressive run (-0.08)

continued on next page

(Table 1 continued from previous page)

Group	Loadings
Passing volume	• long passes (-0.05) • progressive passes (-0.05)
Dueling	• defensive duels (-0.44) • offensive duels won (-0.06) • offensive duels (-0.04)

APPENDIX C

SQL extraction and construction queries

This appendix sketches the logic of the query used to extract and assemble the staged Wyscout data underlying the player–season dataset analysed in this master’s thesis. The real query runs to several hundred lines, dominated by long, mechanically repeated column lists (one line per raw statistic, one per per-90 feature, one per efficiency ratio); the sketch below reproduces its seven-stage logic without reproducing every column, so that the extraction and aggregation logic can be understood and, in principle, reproduced without reading several hundred near-identical lines. The complete, literal query is kept in the project repository at `appendix/sql/wyscout_extraction_query.sql` for anyone who needs it.

1. Extraction and construction logic

```
-- =====
-- Condensed sketch of the LaLiga player-season extraction and construction
-- query. Reproduces the logic of the seven stages below without the full
-- ~90-column select lists of the actual query, which is kept in this
-- directory (wyscout_extraction_query.sql) for anyone who needs the literal
-- SQL. Grain: one row per player_id x season_id x competition_id.
-- =====

with

-- 1) Competition and season scope -----
target_competitions as (
  select competition_id, competition_name, competition_area_name
  from stg_wyscout__competitions
  where lower(competition_name) like '%laliga%'
    and lower(coalesce(competition_gender, 'male')) = 'male'
    and coalesce(competition_division_level, 1) = 1 -- top flight only
),
target_seasons as (
  select season_id, season_name, competition_id
  from stg_wyscout__seasons
  where competition_id in (select competition_id from target_competitions)
),

-- 2) Match-player rows, restricted to the target competition/seasons -----
match_player_base as (
  select
    mps.player_id, mps.match_id, mps.season_id, mps.competition_id,
    p.player_name, p.role_name,
    mps.total_minutes_on_field,
    -- ~90 raw per-match counting statistics, grouped into blocks:
    -- attacking output (goals, shots, xG, touches in box, ...)
    -- passing and creation (passes, key passes, crosses, ...)
    -- carrying / 1v1 (dribbles, progressive runs, ...)
    -- defence and pressing (interceptions, recoveries, ...)
    -- duels (defensive/offensive/aerial duels, won/lost)
    -- discipline (fouls, cards, losses)
```

```

-- plus a block of goalkeeper-specific markers kept only to drive the
-- exclusion step below (total_gk_saves, total_gk_shots_against,
-- total_gk_clean_sheets, total_gk_exits, total_gk_conceded_goals)
mps.*
from stg_wyscout__matches_players_stats as mps
join target_seasons using (season_id)
join target_competitions using (competition_id)
left join stg_wyscout__players as p using (player_id)
),

-- 3) Goalkeeper exclusion, by behaviour rather than by role label -----
-- Role labels alone are not reliable (outfield players who briefly go in
-- goal, mislabelled positions); goalkeeper-specific activity is not.
outfield_match_player_base as (
  select *
  from match_player_base
  where coalesce(total_gk_shots_against, 0) = 0
    and coalesce(total_gk_saves, 0) = 0
    and coalesce(total_gk_conceded_goals, 0) = 0
    and coalesce(total_gk_clean_sheets, 0) = 0
    and coalesce(total_gk_exits, 0) = 0
),

-- 4) Aggregate match rows to one row per player x season -----
player_season_base as (
  select
    player_id, season_id, competition_id,
    max(player_name) as player_name,
    sum(total_minutes_on_field) as minutes,
    -- one sum() per raw counting statistic carried over from step 2
    -- (goals, assists, passes, duels, cards, ... ~90 columns)
    sum(total_goals) as goals,
    sum(total_passes) as passes,
    sum(total_duels) as duels
    -- ... (remaining raw counts omitted here, summed identically)
  from outfield_match_player_base
  group by player_id, season_id, competition_id
),

-- 5) Minimum-exposure filter -----
-- Drops small-sample player-seasons whose per-90 rates would otherwise be
-- dominated by noise.
player_season_filtered as (
  select *
  from player_season_base
  where minutes >= 600
),

-- 6) Feature construction: per-90 rates and efficiency ratios -----
player_season_features as (
  select
    psf.*,
    -- Per-90 intensity: one term per volume feature (~30 features total),
    -- covering attacking output, passing, carrying, defence, and duels.
    90.0 * psf.goals / nullif(psf.minutes, 0) as goals_p90,
    90.0 * psf.passes / nullif(psf.minutes, 0) as passes_p90,
    90.0 * psf.duels / nullif(psf.minutes, 0) as duels_p90,
    -- ... (remaining per-90 features constructed identically)

    -- Efficiency ratios: successful / attempted, one term per ratio
    -- feature (~18 features total: pass accuracy, dribble success rate,

```

```
-- duel win rates by type, shot conversion, goal-per-shot, ...).
1.0 * psf.successful_passes / nullif(psf.passes, 0) as pass_accuracy,
1.0 * psf.duels_won / nullif(psf.duels, 0) as duel_win_rate
-- ... (remaining ratio features constructed identically)
from player_season_filtered as psf
)

-- 7) Final analytical output: one row per player-season -----
select * from player_season_features;
```

LISTING C.1. Condensed sketch of the LaLiga player-season extraction and construction query. Ellipsis comments mark places where the real query repeats the same pattern across additional columns.

Bibliography

1. Theodore W. Anderson and Herman Rubin, *Statistical inference in factor analysis*, Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability (Jerzy Neyman, ed.), vol. 5, University of California Press, 1956, pp. 111–150.
2. Gianluca Baio and Marta Blangiardo, *Bayesian hierarchical model for the prediction of football results*, Journal of Applied Statistics **37** (2010), no. 2, 253–264.
3. David J. Bartholomew, Martin Knott, and Irini Moustaki, *Latent variable models and factor analysis: A unified approach*, 3 ed., Wiley, 2011.
4. Ingwer Borg and Patrick J. F. Groenen, *Modern multidimensional scaling: Theory and applications*, 2 ed., Springer, 2005.
5. Michael W. Browne and Robert Cudeck, *Alternative ways of assessing model fit*, Testing Structural Equation Models (Kenneth A. Bollen and J. Scott Long, eds.), Sage, 1993, pp. 136–162.
6. Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus A. Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell, *Stan: A probabilistic programming language*, Journal of Statistical Software **76** (2017), no. 1, 1–32.
7. Trevor F. Cox and Michael A. A. Cox, *Multidimensional scaling*, 2 ed., Chapman & Hall/CRC, 2001.
8. Andrew Gelman, *Prior distributions for variance parameters in hierarchical models*, Bayesian Analysis **1** (2006), no. 3, 515–534.
9. Andrew Gelman and Jennifer Hill, *Data analysis using regression and multilevel/hierarchical models*, Cambridge University Press, 2007.
10. John Geweke, *Bayesian treatment of the independent student-t linear model*, Journal of Applied Econometrics **8** (1993), no. S1, S19–S40.
11. John Geweke and Guofu Zhou, *Measuring the pricing error of the arbitrage pricing theory*, Review of Financial Studies **9** (1996), no. 2, 557–587.
12. John L. Horn, *A rationale and test for the number of factors in factor analysis*, Psychometrika **30** (1965), no. 2, 179–185.
13. Robert I. Jennrich and Paul F. Sampson, *Rotation for simple loadings*, Psychometrika **31** (1966), no. 3, 313–323.
14. Ian T. Jolliffe, *Principal component analysis*, 2 ed., Springer, 2002.
15. Miguel A. Juárez and Mark F. J. Steel, *Model-based clustering of non-Gaussian panel data based on skew-t distributions*, Journal of the American Statistical Association **105** (2010), no. 490, 298–309.
16. Kenneth L. Lange, Roderick J. A. Little, and Jeremy M. G. Taylor, *Robust statistical modeling using the t distribution*, Journal of the American Statistical Association **84** (1989), no. 408, 881–896.
17. Hedibert Freitas Lopes and Mike West, *Bayesian model assessment in factor analysis*, Statistica Sinica **14** (2004), no. 1, 41–67.
18. Man-Suk Oh and Adrian E. Raftery, *Bayesian multidimensional scaling and choice of dimension*, Journal of the American Statistical Association **96** (2001), no. 455, 1031–1044.
19. Deniz Can Oruç, Lorenzo Cascioli, Luca Stradiotti, Maaïke Van Roy, Pieter Robberechts, and Jesse Davis, *An analysis of the influence of game context on team playing style*, International Workshop on Machine Learning and Data Mining for Sports Analytics, Springer, 2024, pp. 57–68.
20. Gabriel G. Ribeiro, Lilia C. C. Da Costa, Paulo H. Ferreira, and Diego C. do Nascimento, *A bayesian approach to predict performance in football: a case study*, Frontiers in Sports and Active Living **7** (2025), 1486928.
21. Gideon Schwarz, *Estimating the dimension of a model*, The Annals of Statistics **6** (1978), no. 2, 461–464.
22. Patrick E. Shrout and Joseph L. Fleiss, *Intraclass correlations: Uses in assessing rater reliability*, Psychological Bulletin **86** (1979), no. 2, 420–428.
23. Paul A. Swinton, Ben Stephens Hemingway, Iain J. Gallagher, and Eimear Dolan, *Bayesian approaches to quantifying uncertainty in sport and exercise measurements: A guide for practitioners and applied researchers*, SportRxiv preprint, 2023.

24. Jose E. Teixeira, Eduardo Maio, Pedro Afonso, Samuel Encarnacao, Guilherme F. Machado, Ryland Morgans, Tiago M. Barbosa, António M. Monteiro, Pedro Forte, Ricardo Ferraz, et al., *Mapping football tactical behavior and collective dynamics with artificial intelligence: a systematic review*, *Frontiers in Sports and Active Living* **7** (2025), 1569155.
25. Warren S. Torgerson, *Theory and methods of scaling*, Wiley, 1958.
26. Ledyard R. Tucker and Charles Lewis, *A reliability coefficient for maximum likelihood factor analysis*, *Psychometrika* **38** (1973), no. 1, 1–10.
27. Wyscout, *Wyscout glossary*, <https://dataglossary.wyscout.com/>, n.d., Official glossary describing the Wyscout data model used across the platform, reports, and API. Accessed: 2026-04-21.
28. Anna K. Yanchenko and Peter D. Hoff, *Hierarchical multidimensional scaling for the comparison of musical performance styles*, *The Annals of Applied Statistics* **14** (2020), no. 4, 1581–1603.